Row-Level Structural Integrity Diagnostics for Synthetic Tabular Data

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Abstract

Purpose: Synthetic tabular data is widely used in privacy-sensitive settings, but aggregate fidelity metrics can mask record-level joint violations. We introduce the Hybrid Integrity Framework (HIF), a row-level diagnostic tool for auditing whether synthetic rows are structurally consistent with the real data-generating process.

Methods: HIF combines conditional sentinel classifiers for categorical dependencies, continuity-based residual checks for numerical features, and mined implication rules for hard constraints. These components are aggregated using a multiplicative score that strongly limits compensation between components, penalizing joint inconsistencies while retaining rare but valid cases.

Results: Across four generators and five benchmark datasets, HIF exposes dependency violations that are not revealed by marginal metrics, localizes the offending rows, and uncovers a label-conditional utility effect: integrity filtering improves downstream predictive performance only when the prediction target is among the auditor's dependency hubs (a necessary condition), while remaining utility-neutral otherwise.

Conclusion: HIF provides a practical row-level integrity signal for synthetic tabular data and a checkable precondition for when filtering is likely to help. By auditing the joint structure rather than the marginal fit alone, it supports a more reliable downstream use of synthetic data.

Keywords: Synthetic tabular data; Data integrity; Row-level diagnostics; Dependency auditing; Conditional dependencies; Generative models

1 Introduction

Synthetic tabular data are increasingly used in privacy-sensitive domains such as economics [1], finance, and healthcare [2, 3]. Generative models can often closely match per-column marginals, but this does not guarantee that the resulting rows are jointly consistent with the real data-generating process. Therefore, a row can appear individually plausible while remaining structurally inconsistent with the underlying distribution. This is known as the dependency gap. A synthetic patient may combine an age of 14 years with a doctoral degree, and a synthetic transaction may pair a negative quantity with a positive total. Each value can be individually plausible under its marginal distribution, whereas the row remains impossible in the real world.

Standard evaluation metrics, such as KS, TVD, and JCD, are designed to assess aggregate distributional alignment and are therefore largely blind to this record-level failure mode. Explaining and detecting such violations require auditing conditional relationships at the row level. This is particularly important because a model can achieve strong marginal fidelity while generating structurally invalid records. The issue is not only the modeling quality but also a mismatch between the objective of generation and the structure of the real data.

Recent work has highlighted this gap in synthetic data evaluation [4], but most existing metrics remain cohort-level summaries or generic, outlier scores. Isolation Forest and LOF [5, 6] detect geometric anomalies; however, they miss dependency violations in which each feature value is individually plausible. Learned-density baselines can help in some constrained settings, but their effectiveness depends strongly on the encoding and dimensionality of the problem (Section 4.5). Domain-specific rule audits require manually specified constraints, which limit their scalability to new datasets. Therefore, a practical solution needs to discover audit constraints from the data itself, score records individually, and remain tractable across diverse tabular domains.

We present the Hybrid Integrity Framework (HIF), a row-level audit framework for synthetic tabular data. The HIF combines three complementary checks: conditional classifiers for categorical dependencies, continuity-based checks for numeric features, and mined implication rules for hard constraints. These are aggregated through a multiplicative score that strongly limits compensation between components, flagging records whose joint configuration is inconsistent with the conditional structure of the real data. Crucially, the framework learns the relevant dependency structure from the data rather than requiring a handcrafted specification for each domain.

This study makes two main contributions.

- A row-level structural diagnostic (HIF) that evaluates record-level joint conditional integrity, not just aggregate marginals (Section 3).
- A characterization of when integrity filtering improves downstream utility: only when the prediction target is among the auditor's dependency hubs (up to +21.4 F1, $p < 0.001$), otherwise utility-neutral ($p = 0.90$), established via two controlled experiments (Section 4.4).

The remainder of this paper is structured as follows. Section 2 places HIF in the context of prior work on synthetic tabular data generation and evaluation. Section 3 formalizes the proposed methodology. Section 4 presents the empirical

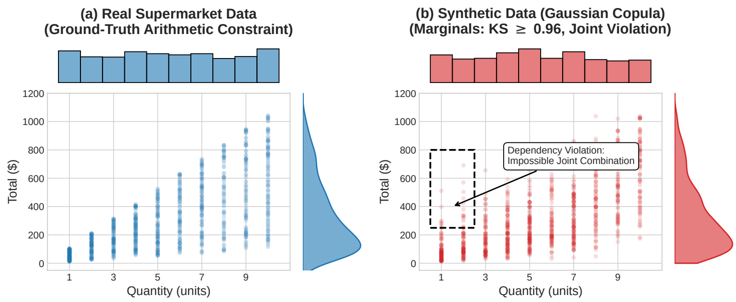

Fig. 1 An empirical demonstration of the Dependency Gap on the Supermarket Sales dataset. While generative models (e.g., Gaussian Copula) successfully replicate the 1D marginal distributions of *Quantity* and *Total* (side histograms; per-column KS ≈ 0.96 for these two marginals, versus the dataset-level KS of 0.903 ± 0.006 reported in Table 1), they fail to preserve the joint arithmetic constraint (Total $=$ Price $\times$ Quantity $\times$ 1.05, where tax is fixed at 5%), hallucinating impossible records (dashed box) that aggregate marginal metrics do not capture.

findings for the benchmarks and generators. Section 5 discusses the implications, limitations, and directions for future work. The complete open-source implementation, benchmark scripts, and dataset configurations are available at https://github.com/ahmed-fouad-lagha/tabular-polygraph.

2 Related Work

Synthetic Tabular Data Generation.

The field has evolved from early copula-based methods [7, 8] to deep-generative architectures. GAN-based models, such as CTGAN and TVAE [9] set an early benchmark for capturing joint distributions by building on the Synthetic Data Vault framework [10], which was later improved for skewed distributions by CTAB-GAN [11]. Diffusion paradigms, such as TabDDPM [12], flow-based gradient-boosted tree generators [13], and composable architectures, such as CoDi [14] have pushed the frontier of manifold matching. Permutation-invariant objectives [15] and federated frameworks [16] extend the synthesis to column-order robustness and decentralized settings. However, both statistical and deep architectures suffer from a fundamental objective mismatch: statistical models excel at per-column marginal fitting but smooth over complex nonlinear conditional constraints, whereas neural models (such as CTGAN) can learn nonlinear categorical contexts but remain prone to mode collapse or localized joint violations. Our work evaluates a representative spectrum of generators, from classical copulas to neural architectures, and shows that the Dependency Gap recurs across these diverse generative paradigms.

Aggregate Evaluation Metrics.

Traditional evaluation pipelines rely on statistical fidelity (Kolmogorov-Smirnov tests, Joint Correlation Distance) and predictive utility (Train-on-Synthetic, Test-on-Real; TSTR). These metrics evaluate aggregate column-level statistics and are blind to row-level conditional dependency violations, a corruption that samples each feature from its correct marginal but ignores inter-feature correlations is entirely invisible to such metrics. Long et al. [4] formalized these "Dependency Gaps" and proposed rule-based audits for inter-column relationships. Yu et al. [17] introduced the SHAP Distance, an explainability-aware fidelity metric, but it yields a dataset-level attribution summary rather than per-record joint-conditional scores and scales exponentially with feature count (Appendix D.3). Distribution-level auditing faces provable sample-complexity limits: identity and uniformity tests require sample sizes scaling with domain support [18–20], which is partially mitigated for product-structured distributions [21], and conditional independence testing is sample-intensive in high dimensions [22]. These bounds motivate the individual scoring of records rather than the global testing of the synthetic cohort. We quantify these limits in the HIF operating configuration in Appendix D.2.

Generation-Side Constraint Enforcement.

Umesh et al. [23] studied the preservation of logical and functional dependencies from a generation-side perspective, constraining the generative model to respect mined dependencies at the training time. This is complementary to, but distinct from, our post-hoc audit of arbitrary generator outputs.

Record-Level Auditing.

A complementary line scores individual records rather than a synthetic cohort. Alaa et al. [24] introduced sample-level support metrics (α-Precision, β-Recall) auditing whether a record falls within the support of the real distribution in a learned embedding space—a neighborhood-membership notion rather than a check of conditional inter-feature logic, with hand-selected embeddings. Data-SUITE [25] identifies in-distribution incongruous examples through per-feature conformal prediction intervals, but scores each feature marginally, targets real samples, and does not evaluate joint conditional configurations. DOMIAS [26] performs per-record auditing via membership inference against overfitted generators, which is a privacy signal, not a general structural integrity check. Two recent studies merit an explicit distinction. TabStruct [27] benchmarks structural fidelity across 13 generators and 29 datasets, but its scoring is dataset-level and requires (or approximates) ground-truth causal structure, whereas HIF scores each record against conditional dependencies mined from the data itself, including implication rules discovered in an Apriori-style. Curated LLM [28] curate candidate synthetic rows in low-data regimes using a supervised signal relying on the learning dynamics of a model trained on real labels; this is the label-conditioned regime isolated in Section 4.4, whereas HIF operates without downstream labels and derives a utility consequence only as an emergent, label-conditional property. To our knowledge, no prior work composes mined functional dependencies and implication rules into a per-record integrity audit of an arbitrary generator's output; the discovery literature for functional dependencies and denial constraints [29, 30] has largely focused on schema design, data cleaning, and constraint mining rather than row-level auditing of synthetic tabular data. This constraint-mining bridge is the key distinction between HIF and prior work in this field.

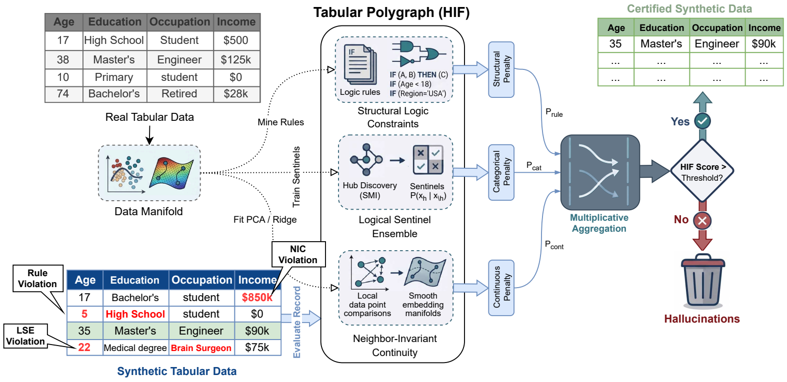

Fig. 2 Architecture of the Hybrid Integrity Framework (HIF). Synthetic tabular data is verified by the Logical Sentinel Ensemble (LSE) and Neighbor-Invariant Continuity (NIC) modules, combined via multiplicative aggregation to filter semantically inconsistent records.

3 Mathematical Foundation

The Hybrid Integrity Framework (HIF) formalizes synthetic data auditing as a probabilistic conditional scoring task. Central to this task is the distinction between four related concepts: *fidelity*—aggregate distributional alignment, as measured by marginal metrics such as KS, TVD, and JCD; *integrity*—the row-level property HIF audits, where a record has high integrity if its joint configuration is consistent with the conditional dependencies learned from the real data; *plausibility*—a weaker, per-column property meaning each value falls within the expected marginal range, which a dependency-violating record can still satisfy; and *validity*—whether a record is actually possible under the true data-generating process. Because HIF certifies consistency with a *learned* conditional manifold, a low-integrity record is best read as being "structurally inconsistent with the training distribution" rather than universally invalid (Section 4).

Let $\mathcal{D}_{real}$ denote the original training dataset. We define the **ground-truth manifold** $\mathcal{M}_{real}$ as the support of the joint distribution $P_{real}(\mathbf{x})$, which is the localized subspace of valid feature configurations that occur in reality [31]. A synthetic record $\mathbf{x} = [\mathbf{x}_{cat}, \mathbf{x}_{cont}]$ (with $\mathbf{x}_{cat}$ and $\mathbf{x}_{cont}$ being the sub-vectors of categorical and continuous features, respectively) is a *dependency-violating record* if it is plausible under each marginal distribution yet falls outside the ground-truth manifold (i.e., it is highly improbable under the joint conditional distribution $P_{\mathcal{D}_{real}}(\mathbf{x}_h \mid \mathbf{x}_{\setminus h})$ learned from the data). This distinction is central: aggregate fidelity metrics (KS, TVD, and JCD) assess whether records fall within the expected marginal range of each column in isolation, whereas HIF additionally evaluates whether the joint configuration of each record structurally aligns with the ground-truth manifold. Figure 2 shows how the three auditing layers compose into a single row-level score, and the complete calibration process for establishing these data-driven manifold boundaries is detailed in Algorithm 2.

Algorithm 1 Hybrid Integrity Framework (HIF) — End-to-End Audit

Input: Real data $\mathcal{D}_{real}$, synthetic record $\mathbf{x}$, hub count K
Output: Integrity score $H(\mathbf{x}) \in [0, 1]$
Calibration (on $\mathcal{D}_{real}$):
$\mathcal{H} \leftarrow$ top-K features by Feature Dependency Score $S(h)$
for $h \in \mathcal{H}$ **do**
    Train Sentinel f_h (Random Forest) to predict x_h from $\mathbf{x}_{\setminus h}$
    $\delta_h \leftarrow \max(Q_{0.05}(\text{OOB probabilities}), 0.01)$ ▷ Confidence floor
end for
$\mathcal{Z} \leftarrow \text{SVD}(\text{Scale}(\text{OneHot}(\mathcal{D}_{cat})))$ ▷ Spectral manifold
for continuous feature y **do**
    Train Gradient Boosting regressor $r_y : \mathcal{Z} \to \mathbb{R}$
    $\tau_y \leftarrow \max(Q_{0.95}(\mathcal{E}_y), \text{Median}(\mathcal{E}_y) + 2\text{MAD}(\mathcal{E}_y))$
    $\gamma_y \leftarrow \max(Q_{0.98}(\mathcal{E}_y), 2\tau_y, 3\text{MAD}(\mathcal{E}_y), 0.1)$
end for
$\mathcal{R} \leftarrow \text{MineImplicationRules}(\mathcal{D}_{real})$ ▷ Apriori, min_conf=0.95
Scoring (for each synthetic x):
for $h \in \mathcal{H}$ **do**
    $\mathcal{P}_h \leftarrow \text{clip}((\delta_h - f_h(\mathbf{x}_h \mid \mathbf{x}_{\setminus h}))/\delta_h, 0, 1)$
end for
$\mathcal{P}_{cat} \leftarrow 1 - \prod_{h \in \mathcal{H}} (1 - \mathcal{P}_h)$
$\mathbf{z} \leftarrow \phi(\mathbf{x}_{cat})$
for continuous feature y **do**
    $\mathcal{P}_y \leftarrow \text{clip}((|y - r_y(\mathbf{z})| - \tau_y)/\gamma_y, 0, 1)$
end for
$\mathcal{P}_{cont} \leftarrow \max_y \mathcal{P}_y$
$\mathcal{P}_{rule} \leftarrow \mathbb{I}[\mathbf{x} \text{ violates any } r \in \mathcal{R}]$
$H(\mathbf{x}) \leftarrow ((1 - \mathcal{P}_{cat})(1 - \mathcal{P}_{cont})(1 - \mathcal{P}_{rule}))^{1/3}$

3.1 Logical Sentinel Ensemble (LSE)

To audit categorical consistency, we first identify the set of Dependency Hubs $\mathcal{H}$ that govern the structural dependencies of the dataset. For a given row $\mathbf{x}$, let $x_h \in \mathcal{X}_h$ denote the categorical value of target feature h, and let $\mathbf{x}_{\setminus h}$ denote the values of all other features (i.e., $\mathbf{x}$ with feature h removed).

We train an independent Random Forest classifier [32]—leveraging decision tree recursive partitioning to capture non-linear feature interactions [33]—which we call a *Sentinel* and denote as f_h, to predict x_h given $\mathbf{x}_{\setminus h}$. The sentinel outputs a conditional probability distribution $P_{f_h}(x_h = c \mid \mathbf{x}_{\setminus h})$ for all possible categories $c \in \mathcal{X}_h$. We define the importance of a feature h using the Feature Dependency Score $S(h)$, the cross-validated classification accuracy of its sentinel:

$$S(h) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{real}} \left[\mathbb{I} \left(\arg\max_{c \in \mathcal{X}_h} P_{f_h}(x_h = c \mid \mathbf{x}_{\setminus h}) = x_h \right) \right] \tag{1}$$

where $\mathbb{I}$ is an indicator function. By maximizing $S(h)$, we selected the hubs that were most strongly constrained by conditional interactions with other attributes. While computationally efficient, this reliance on single-variable classification accuracy primarily captures univariate dependencies and may underrepresent higher-order parity relationships. Future work will investigate Total Correlation and multi-way interaction kernels to broaden hub selection.

Consider, for illustration, a demographic dataset where x_h is *Education Level* and $\mathbf{x}_{\setminus h}$ contains *Age* (e.g., 14 years). Because a 14-year-old cannot hold a PhD in the real world, the sentinel f_h learns from the ground-truth data to assign $P_{f_h}(\text{PhD} \mid \text{Age} = 14) \approx 0$. If a generative model hallucinates a 14-year-old with a PhD, the sentinel flags the record because the assigned probability for the hallucinated category is extremely low.

Confidence Floor Calibration.

Drawing inspiration from distribution-free quantile calibration in inductive conformal prediction [34–36], we establish a data-driven Confidence Floor δ_h for each hub based on empirical calibration error bounds, defined as the empirical 5th-percentile ($Q_{0.05}$) of the *out-of-bag* probability mass assigned to the correct category by the sentinel in the ground-truth data, with a safety minimum of 0.01:

$$\delta_h = \max\Big(Q_{0.05}\Big(\{P_{f_h}(x_h \mid \mathbf{x}_{\setminus h}) \mid \mathbf{x} \in \mathcal{D}_{real}\}\Big), 0.01\Big) \tag{2}$$

Using out-of-bag predictions (i.e., predictions of the trees that did not train on each row) yields out-of-sample floors without requiring a held-out split, preventing over-optimistic calibration on rows that the sentinels memorized during training. This 5th-percentile floor acts as a conservative safety margin: if the real data occasionally contain 18-year-olds with college degrees, the sentinel's probability might be low but non-zero (e.g., 0.05), and the floor δ_h ensures that we do not penalize rare but valid records that fall within the natural density of the ground-truth manifold.

The per-hub penalty $\mathcal{P}_h(\mathbf{x})$ uses a soft-threshold activation to suppress noise-level deviations, harshly penalizing only those records whose probabilities fall far below the confidence floor (such as the impossible 14-year-old PhD):

$$\mathcal{P}_h(\mathbf{x}) = \text{clip}\left(\frac{\delta_h - P_{f_h}(x_h \mid \mathbf{x}_{\setminus h})}{\delta_h}, 0, 1\right) \tag{3}$$

The categorical penalty aggregates multiplicatively across hubs, ensuring that ruptures in multiple manifold dimensions are compounded.

$$\mathcal{P}_{cat}(\mathbf{x}) = 1 - \prod_{h \in \mathcal{H}} (1 - \mathcal{P}_h(\mathbf{x})) \tag{4}$$

The complete manifold calibration procedure, including hub selection, confidence floor calibration, NIC regressor training, and rule mining, is summarized in Algorithm 2 (Appendix B).

3.2 Neighbor-Invariant Continuity (NIC)

For continuous features $\mathbf{y}$, we evaluate integrity via Semantic Adjacency. We map the categorical context to a continuous latent space $\mathcal{Z}$ using a standardized spectral projection $\phi : \mathcal{X}_{cat} \to \mathcal{Z}$, drawing on classical Multiple Correspondence Analysis (MCA) and Singular Value Decomposition (SVD) on binary indicator matrices [37, 38]. Standardizing the indicator matrices by their standard deviation prior to SVD mathematically replicates the inverse frequency weighting of the MCA, ensuring that rare but highly informative categorical constraints dominate the continuous manifold projection. We then model the conditional expectation of each continuous feature y as a non-linear function of this manifold using a Gradient Boosting Regressor r_y [39, 40]:

$$\hat{y}(\mathbf{x}) = \mathbb{E}[y \mid \phi(\mathbf{x}_{cat})] \approx r_y(\phi(\mathbf{x}_{cat})) \tag{5}$$

For any given record $\mathbf{x}$, let $e_y(\mathbf{x}) = |y(\mathbf{x}) - \hat{y}(\mathbf{x})|$ denote the absolute regression residual, and define the ground truth residual set as $\mathcal{E}_y = \{e_y(\mathbf{x}) \mid \mathbf{x} \in \mathcal{D}_{real}\}$. For example, a synthetic 45-year-old Senior Executive generated with an income of \$12 yields an enormous residual against the expected executive income ($\hat{y} \approx$ \$120,000), far exceeding the natural variance of executives in the real dataset and signaling a continuous dependency gap. To handle extreme feature skewness (common in census domains), we define a robust hybrid threshold τ_y from $\mathcal{E}_y$ using the Median Absolute Deviation (MAD) [41], a robust estimate of local manifold spread that is less sensitive to outliers than the standard deviation:

$$\tau_y = \max\left(Q_{0.95}(\mathcal{E}_y), \mathrm{Median}(\mathcal{E}_y) + 2 \cdot \mathrm{MAD}(\mathcal{E}_y)\right) \tag{6}$$

This threshold keeps the NIC layer stable, even in high-skew distributions, such as macroeconomic wage data.

Two scope notes bound the interpretation of the NIC layer. First, the threshold conditions on the categorical context only: NIC audits each continuous feature against the learned categorical manifold $\phi(\mathbf{x}_{cat})$, so it cannot by construction detect continuous–continuous dependencies such as the arithmetic identity of Figure 1 (Total = Price $\times$ Quantity $\times$ 1.05 on Supermarket Sales, where tax is fixed at 5%; on Online Purchases tax is a separate column). Such relations are the province of the rule layer, which quantizes and mines them as implication rules, and of the LSE, which treats quantile-binned continuous features as hubs; on the Supermarket Sales audit, the identity is caught by those layers while the NIC component rate is 0.0% (Section 4.2). Therefore, NIC is a categorical context continuity check rather than a general multivariate residual auditor. Second, τ_y is calibrated on the residuals of the training rows themselves, and because the regressor is fitted on those same rows, the in-sample residuals can modestly understate the natural noise floor. A held-out 50/50 re-calibration across our five benchmarks yields thresholds between 0.96$\times$ and 1.54$\times$ looser than the in-sample thresholds (Census ACS, where the headline analyses sit, $\leq$ 1.05$\times$); therefore, NIC's per-column violation rates are mild upper bounds rather than exact estimates. We retain in-sample calibration because it uses the full real cohort, and the effect is second-order relative to the LSE signal; the sensitivity is disclosed rather than silently propagated.

The continuous penalty $\mathcal{P}_{cont}(\mathbf{x})$ uses a dynamic scaling factor $\gamma_y = \max(Q_{0.98}(\mathcal{E}_y), 2\tau_y, 3\text{MAD}(\mathcal{E}_y), 0.1)$ that bounds the penalty relative to the feature's natural noise floor, so small deviations beyond τ_y yield soft penalties, whereas massive violations (like the \$12 executive) yield a maximum penalty of 1.0:

$$\mathcal{P}_{cont}(\mathbf{x}) = \max_y \left[\text{clip} \left(\frac{e_y(\mathbf{x}) - \tau_y}{\gamma_y}, 0, 1 \right) \right] \tag{7}$$

As a defensive guard, NIC detects *collapsed* continuous features whose conditional predictions have near-zero variance (i.e., the feature is effectively constant given the categorical context); for such degenerate features, all records receive a uniform base penalty because the expected residual distribution provides no reliable baseline. This guard was not activated on any benchmark dataset in our experiments.

3.3 Multiplicative Aggregation

In contrast to additive fidelity metrics that allow a model to "compensate" for logical failures with high statistical overlap, HIF formalizes integrity as a multiplicative conjunction: the hybrid integrity score $H(\mathbf{x})$ is defined as the geometric mean of the component validities:

$$H(\mathbf{x}) = \left(\prod_{l \in \{cat, cont, rule\}} (1 - \mathcal{P}_l(\mathbf{x})) \right)^{1/L} \tag{8}$$

where l iterates over active auditing layers. The geometric mean strongly limits compensation compared to additive aggregation: a significant failure in any single manifold dimension ($\mathcal{P}_l \to 1$) drives the entire record integrity toward zero, regardless of how well the other components align (see Table 10, Section 4.7). Unlike a strict Logical AND, the geometric mean can still partially rescue a low component score if other components are near-perfect, but this rescuing effect is far weaker than under arithmetic averaging. For a formal analysis of this aggregation logic under explicit assumptions—intersection soundness (Proposition 1), asymptotic convergence (Proposition 2), and the Type I false rejection bound under quantile calibration (Remark 1)—we direct the reader to **Appendix A**.

While HIF prioritizes soft probabilistic validities via LSE and NIC, we incorporate symbolic implication rules as an optional *Structural Logic Constraint* layer ($l = rule$) to enforce known deterministic or physical laws (e.g., Age $\geq$ 18 for voters), bridging differentiable manifolds and rigid symbolic constraints. Rules are mined from the real data with minimum confidence 0.95, minimum support 0.005, at most two antecedents, and up to 25 rules, and they are applied as hard binary penalties: a record violates the layer if any mined rule fails. Component validities are floored at $\epsilon = 10^{-4}$, so a rule-violating record attains $H(\mathbf{x}) \leq \epsilon^{1/L} \approx 0.046$ in the three-layer configuration, safely below the 0.5 frontier.

The Integrity Frontier.

The dataset-level HIF score is the expected record-level integrity score. To provide a discrete diagnostic, a record is classified as a *Semantic Violation* if $H(\mathbf{x}) < 0.5$, defining the framework Integrity Frontier.

$$\text{HIF Score} = \frac{1}{N}\sum_{i=1}^{N} H(\mathbf{x}_i) \tag{9}$$

This score provides a bounded diagnostic $\in [0, 1]$, where 1.0 indicates a logically certified synthetic cohort. The complete end-to-end auditing procedure for scoring a synthetic dataset is detailed in Algorithm 3 (Appendix B).

Because HIF penalizes records that violate *learned conditional dependencies* rather than marginal rarity—the confidence floor δ_h is calibrated on the ground-truth manifold precisely to avoid penalizing rare-but-valid tail records—filtering operates on structural consistency rather than demographic support; a systematic audit of representation drift across sensitive attributes under integrity filtering is left to future work.

4 Results

We evaluated HIF on five benchmark datasets with diverse structural complexities: U.S. Census ACS, Adult Income, Credit Card Default, Online Purchases, and Supermarket Sales. Unless otherwise stated, we report the mean ± standard deviation across $N = 3$–10 random seeds, depending on the experiment.

4.1 Experimental Setup

We compared HIF against four aggregate fidelity metrics: KS complement, TVD complement, Moment Matching, and Joint Correlation Distance. We also report α-precision and β-recall as support-overlap diagnostics. The controlled corruption protocol shuffles values within strongly constrained dependencies to preserve marginals while breaking the row-level conditional structure; this isolates the Dependency Gap from ordinary marginal drift. For baseline comparability, the Isolation Forest and LOF were evaluated at a known contamination rate, and the learned density baseline used a matched real-data score threshold. We report both threshold-independent ranking metrics (ROC-AUC and PR-AUC) and a fixed-threshold F1.

Compute and software. All experiments were run on commodity multi-core hardware (AMD EPYC 7742, 64 cores) with a single NVIDIA A100 40GB GPU for LightGBM training. The total compute time for the full benchmark suite (5 datasets $\times$ 4 generators $\times$ 10 seeds) was approximately 12 GPU-hours. Software: Python 3.10, scikit-learn 1.3.2, LightGBM 4.1.0, NumPy 1.24.3, and SciPy 1.11.1. Random seeds were fixed per configuration; all results report mean ± standard deviation across $N = 3$–10 seeds as stated.

The main empirical question is not whether HIF outperforms marginal metrics in a trivial sense, but whether it exposes dependency violations that aggregate statistics. The answer is affirmative across all benchmark suites.

Main findings.

First, aggregate fidelity does not imply a structural integrity. Second, HIF localizes genuine dependency violations that are externally verifiable against arithmetic constraints in real data. Third, the utility gains from filtering are label-conditional: they emerge when the prediction target is among the auditor's selected hubs, and they vanish otherwise. Fourth, HIF is competitive with and in several settings stronger than standard anomaly baselines on conditional dependency failures. Finally, multiplicative aggregation that strongly limits compensation is essential to the framework's behavior.

4.2 Cross-Architecture Maturity Audit and Cohort Certification

To evaluate the diagnostic utility of HIF across generative paradigms, we conducted a cross-architecture audit across four benchmark datasets from diverse domains (Table 1): Supermarket Sales (retail arithmetic constraints), Online Purchases (e-commerce transaction constraints), Credit Card Default (financial classification), and Adult Census Income (demographic benchmark). The results reveal a key insight: an *Integrity-Fidelity Decoupling*: a generator can match aggregate marginal statistics (KS) while simultaneously violating the joint conditional structure of the data, such that high marginal fidelity does not imply row-level integrity, and vice versa.

1. Integrity-Fidelity Decoupling on Constrained Domains.

Across all four datasets, the Vine Copula arm achieved the highest marginal statistical fit of the four generators (KS $\geq$ 0.970). This arm is a designed positive control: its categorical columns are sampled independently from their marginal distributions, and its categorical–numeric dependence is deliberately not modelled (as disclosed in the released implementation), and its near-perfect KS is the expected consequence of resampling the empirical marginals. Therefore, it maximizes the contrast between the aggregate marginal fidelity and row-level joint integrity by construction. Accordingly, on datasets with cross-feature relational constraints (Supermarket Sales and Online Purchases), it exhibits severe joint dependency failures, with violation rates of **85.5% to 99.9%** (HIF $\leq$ 0.271); CTGAN and TVAE show similar deficits on transaction constraints (HIF = 0.02–0.30, violation rates 81.3%–100%). Marginal metrics (KS) report these cohorts as high-fidelity, underscoring that aggregate fidelity metrics do not capture the joint dependency ruptures. A qualitative audit on Supermarket Sales (Vine Copula) confirmed the mechanism: generators match each column's marginal distribution yet violate the arithmetic identity subtotal = unit_price $\times$ quantity $\times$ 1.05 (tax fixed at 5%), with HIF flagging essentially the entire cohort (violation rate 99.9% across $N = 10$ seeds; component rates 100.0% LSE, 59.3% rules, 0.0% NIC)while KS and joint-correlation metrics reported high fidelity. This identity is externally verifiable, independent of the auditor's learned models, and anchors the construct validity of the HIF penalty in a way that learned conditional improbability alone cannot (see the threats-to-validity discussion in Section 4).

These violations are confirmed against arithmetic identities that hold to $\approx 10^{-13}$ relative error in the real data—Total = Unit Price $\times$ Quantity $\times$ 1.05 (tax fixed at 5%;

equivalently Total = cogs + gross income) on Supermarket Sales, and item_total = item_subtotal + item_tax together with item_subtotal = purchase_price × quantity on Online Purchases—we verified every real generator cohort in this audit ($N = 10$ seeds) using these arithmetic identities. Records flagged by HIF ($H < 0.5$) are confirmed as identity-violating in $\geq$ 98.9% of cases across all eight dataset–generator configurations (1.000 in six of eight; Table 2). This verification re-fits HIF on the projection onto the five identity-bearing columns, where the arithmetic identity becomes a near-deterministic hub; therefore, this confirmation is not fully independent of the auditor's learned models. The flagged rate in this audit is not directly comparable to the full-feature Viol. Rate of Table 1—the projection isolates the constraint and thus flags more rows (e.g., 32.0% vs. 3.4% for Gaussian Copula on Supermarket Sales), which is precisely why it serves as the sharper external anchor for the penalty. The constraints are violated wholesale (base violation rate 0.87–1.00), so on these fully constrained domains, the identity check confirms the Dependency Gap rather than discriminating among records: the cleanest discriminator is Gaussian Copula on Online Purchases, where unflagged records violate at 0.715 versus 0.989 among flagged, and the HIF penalty correlates with violation severity (Spearman $\rho = 0.46 \pm 0.01$; Table 2). Conversely, where HIF flags essentially the entire cohort (Vine and CTGAN on Supermarket Sales; Vine, CTGAN, and TVAE on Online Purchases), its penalty is only weakly monotone in arithmetic severity ($\rho \leq 0.44$), because the cohort's failures are dominated by categorical-structure violations rather than continuous residuals, precisely the regime in which the learned-density baseline of Section 4.5 is the sharper detector of the arithmetic constraint (GMM ROC-AUC 0.955 vs. 0.797 on Online Purchases). These numbers quantify the earlier qualitative audit: the flagged cohort is externally invalid at near-universal rates, while HIF's *ranking* within arithmetic-constraint domains is limited—a boundary that motivates the complementary learned-density baseline reported in Section 4.5.

2. Evaluation on Unconstrained Manifolds.

On unconstrained demographic and financial benchmarks (Adult Income, Credit Default), raw generators preserve conditional structures better, with HIF scores ranging from 0.31 to 0.99 (violation rates 0.2%–70.3%), providing a continuous gradient of structural integrity for comparing generator maturity. Notably, TVAE achieved the highest HIF on both Credit Default (0.908) and Adult Income (0.990). Read against the auditor's own reference on genuine held-out real data (Table 1): Credit Default real rows score HIF 0.823 with a 14.0% violation rate, and Adult real rows score 0.923 with 4.3% violation rate. The TVAE on Adult (0.990, 0.2% violations) therefore exceeds the real-data reference on both quantities, which is evidence of genuine joint-dependency preservation. TVAE on Credit (0.908, 8.0%) likewise exceeds it (0.823, 14.0%); thus, in these runs, TVAE's joint-dependency preservation of TVAE generalizes beyond the auditor's own training distribution rather than reproducing its estimation-error floor. The auditor also scores its training data higher than fresh held-out real rows (Section 4), a framework-level overfitting signature that argues for reading HIF values relative to the real-data reference rather than absolutely.

3. Quantitative Decoupling Evidence.

Pearson correlation analysis at the per-seed level across the four audit benchmarks ($n = 160$: 16 dataset–generator configurations × 10 seeds, with Census ACS held out as the calibration benchmark; Table 1) precisely characterizes the decoupling. Seeds within a configuration are not independent observations; therefore, these correlations are descriptive rather than inferential: HIF shows no linear relationship with marginal fidelity ($\rho(\mathrm{HIF}, \mathrm{KS}) = 0.094$) or support coverage ($\rho(\mathrm{HIF}, \alpha) = 0.06$), but a strong positive correlation with joint correlation fidelity ($\rho(\mathrm{HIF}, \mathrm{JCD}) = +0.83$). Thus, HIF aligns with the joint-dependency structure while being decoupled from per-column marginal metrics: high marginal fidelity (KS) does not imply high integrity, exactly the pattern observed for Vine Copulas on constrained domains.

Table 1 Cross-Architecture Audit: Statistical Fidelity (KS), Support Overlap (α-Precision, β-Recall), and Joint Dependency Integrity (HIF) across benchmark datasets ($N = 10$ seeds, Mean ± SD). High marginal fidelity (KS > 0.96) or high support coverage (α-Prec > 0.79) can fail to imply joint conditional integrity. On datasets with cross-feature constraints, the models exhibited up to 100% dependency violation rates. The "Real data (held-out)" row per dataset is a single deterministic reference: the auditor fitted on real rows and scored on a genuine held-out remainder (see Section 4), establishing the framework's own error floor—genuine real data is flagged at 0.0–14.0% across domains.

Generator	KS (↑)	α-Prec (↑)	β-Rec (↑)	HIF (↑)	Viol. Rate (↓)
Supermarket Sales					
Real data (held-out)	—	—	—	0.980	0.0%
Gaussian Copula	0.903 ± 0.006	0.892 ± 0.012	0.918 ± 0.013	**0.955 ± 0.008**	**3.4 ± 0.8%**
Vine Copula	**0.975 ± 0.002**	0.794 ± 0.008	0.866 ± 0.015	0.028 ± 0.005	99.9 ± 0.1%
CTGAN	0.785 ± 0.034	0.628 ± 0.069	0.878 ± 0.020	0.015 ± 0.012	100.0 ± 0.0%
TVAE	0.594 ± 0.055	0.523 ± 0.051	0.618 ± 0.073	0.035 ± 0.005	100.0 ± 0.0%
Online Purchases					
Real data (held-out)	—	—	—	0.918	6.8%
Gaussian Copula	0.753 ± 0.010	0.488 ± 0.013	0.822 ± 0.023	**0.637 ± 0.014**	**38.7 ± 2.0%**
Vine Copula	**0.970 ± 0.004**	0.829 ± 0.018	0.952 ± 0.010	0.271 ± 0.007	85.5 ± 1.0%
CTGAN	0.662 ± 0.064	0.524 ± 0.070	0.868 ± 0.068	0.304 ± 0.042	81.3 ± 5.3%
TVAE	0.718 ± 0.018	0.610 ± 0.033	0.487 ± 0.026	0.149 ± 0.012	99.8 ± 0.1%
Credit Default					
Real data (held-out)	—	—	—	0.823	14.0%
Gaussian Copula	0.749 ± 0.004	0.738 ± 0.024	0.951 ± 0.010	0.487 ± 0.081	46.2 ± 9.9%
Vine Copula	**0.982 ± 0.002**	0.801 ± 0.011	0.813 ± 0.013	0.404 ± 0.039	56.2 ± 4.6%
CTGAN	0.728 ± 0.063	0.672 ± 0.106	0.855 ± 0.028	0.305 ± 0.052	70.3 ± 6.7%
TVAE	0.732 ± 0.015	0.515 ± 0.034	0.762 ± 0.032	**0.908 ± 0.053**	**8.0 ± 5.3%**
Adult Income					
Real data (held-out)	—	—	—	0.923	4.3%
Gaussian Copula	0.854 ± 0.006	0.936 ± 0.007	0.836 ± 0.014	0.795 ± 0.044	14.8 ± 5.2%
Vine Copula	**0.982 ± 0.005**	**0.968 ± 0.011**	0.788 ± 0.016	0.716 ± 0.069	22.1 ± 7.8%
CTGAN	0.692 ± 0.085	0.804 ± 0.034	0.808 ± 0.050	0.618 ± 0.053	34.4 ± 5.9%
TVAE	0.751 ± 0.019	0.350 ± 0.065	0.425 ± 0.042	**0.990 ± 0.007**	**0.2 ± 0.4%**

Table 2 External Verification of HIF-Flagged Records against Real-Data Arithmetic Identities ($N = 10$ seeds, mean; $\pm$ denotes SEM). Flagged records ($H < 0.5$) are confirmed as identity-violating at $\geq 98.9\%$ in every configuration, confirming that the HIF penalty targets genuinely invalid structures. The verification re-fits HIF on the projection onto the five identity-bearing columns, where the arithmetic identity becomes a near-deterministic hub; therefore, this confirmation is not fully independent of the auditor's learned models. Where both valid and invalid cohorts coexist (Gaussian Copula), unflagged records are markedly less likely to violate (**99.6**% and **71.5**% confirmed vs. 98.9–100% among flagged) and the HIF penalty tracks identity severity (Spearman **0.54** and **0.46**). The flagged rate in this audit is not directly comparable to the full-feature Viol. Rate of Table 1 (e.g., 32.0% vs. 3.4% for Gaussian Copula on Supermarket Sales, which is fully constrained only in the projection).

Dataset	Generator	Flagged	Base Viol.	Conf. @Flagged	Conf. @Unflagged	Spearman ρ
Supermarket Sales						
	Gaussian Copula	32.0%	99.7%	100.0%	**99.6%**	**0.536 $\pm$ 0.006**
	Vine Copula	99.7%	100.0%	100.0%	100.0%	0.096 $\pm$ 0.010
	CTGAN	99.7%	100.0%	100.0%	100.0%	+0.059 $\pm$ 0.026
	TVAE	91.8%	100.0%	100.0%	100.0%	0.440 $\pm$ 0.032
Online Purchases						
	Gaussian Copula	55.6%	86.7%	98.9%	**71.5%**	**0.461 $\pm$ 0.012**
	Vine Copula	97.3%	100.0%	100.0%	100.0%	0.123 $\pm$ 0.015
	CTGAN	95.1%	99.9%	99.9%	99.5%	0.067 $\pm$ 0.038
	TVAE	99.9%	100.0%	100.0%	100.0%	0.217 $\pm$ 0.018

4.3 Alignment with Downstream Utility

A critical requirement for any synthetic data metric is its ability to act as a proxy for downstream predictive utility, a validation paradigm that is now standard in privacy-sensitive domains [42]. We measured downstream TSTR F1 under integrity filtering across representative generative architectures (Table 3), using a Random Forest classifier [32] (via *Scikit-learn* [43]); the *Rule-only Baseline* prunes records violating any mined implication rules, whereas the *HIF Filtered* variant uses the semantic frontier ($H < 0.5$).

The paired tests on Census ACS ($N = 10$ seeds, binary median-split `household_income` target) quantify the asymmetry. For the Gaussian Copula, filtering yielded $p = 0.90$ (95% CI $[-0.003, +0.003]$): no detectable utility degradation, as expected for a cohort largely free of conditional dependency violations (violation rate 1.3%). For CTGAN, downstream utility gains were statistically significant (paired t-test $p = 0.008$; Wilcoxon $p = 0.002$; 95% CI $[+0.020, +0.099]$; paired Cohen's $d \approx 1.08$), with F1 improving from 0.381 ± 0.063 to 0.441 ± 0.096 ($+0.060$), supporting the claim that HIF is particularly useful for architectures that produce joint dependency violations. The *Rule-only Baseline* isolates the symbolic layer: pruning the records that violate mined implication rules ($\approx 7\%$ of the cohort; retention 92.9%) leaves utility essentially unchanged (Census ACS CTGAN: F1 0.367 vs. 0.381 unfiltered; Gaussian Copula: 0.941 vs. 0.942), whereas full HIF filtering raises CTGAN F1 to 0.441 at 63.0% retention. Therefore, utility recovery is driven by the LSE/NIC conditional-dependency layers rather than by hard-rule pruning alone. On Online Purchases with the Gaussian Copula (39% violation rate), filtering retains 61.3% of records and F1 moves within noise ($0.977 \rightarrow 0.980$), tempering the hypothesis that dependency violations always act as predictive noise: whether removing them improves utility depends on the strength of the violation-to-label association.

The utility-recovery conclusion is robust to the operating threshold, not merely to the continuous score (Table D1). Sweeping the retention frontier $H \geq \{0.7, 0.5, 0.3\}$ on the Census ACS CTGAN cohort ($N = 10$ seeds) yields F1 scores of 0.761, 0.441, and 0.382 at 30.0%, 63.0%, and 88.5% retention (unfiltered 0.381). The recovery effect is significant at the strict and default frontiers (paired t-test $p < 0.001$ and $p = 0.008$; Wilcoxon $p = 0.002$; 95% CIs $[+0.332, +0.428]$ and $[+0.020, +0.099]$) and increases monotonically with filtering strictness, but disappears at the loose $H \geq 0.3$ frontier ($p = 0.92$), where only the most egregious records are removed.

Protocol disclosure. As is standard in TSTR [42], we fit each generator on the full real cohort and evaluate the downstream classifier on a 30% real holdout carved from that same cohort; the synthetic training distribution therefore overlaps the real rows used as test data, and absolute TSTR levels are optimistically biased. This does not affect the paired filtering comparisons reported below, since both arms share the identical synthetic cohort and the ΔF1 differences are protected from the leak; the cross-architecture conclusions of Section 4.2 likewise do not rely on TSTR levels. We disclose the protocol rather than present absolute TSTR levels as an unbiased estimate.

Table 3 Downstream utility filtering across all configurations (Mean ± SD, $N = 10$ seeds). The retained cohort size (retention) dictates whether the utility can be reliably estimated. In every configuration with a significant gain, the median-split target is itself one of the auditor's selected hubs (`household_income` for Census ACS, `item_total` for Online Purchases); Section 4.4 shows this to be the governing condition.

Dataset	Generator	Retention (%)	F1 (Full)	F1 (HIF Filtered)	Δ F1
Adult	CTGAN	65.6	0.444 ± 0.016	0.451 ± 0.018	+0.007
	Gaussian Copula	85.2	0.599 ± 0.025	0.601 ± 0.033	+0.002
	TVAE	99.8	0.499 ± 0.067	0.499 ± 0.067	+0.001
	Vine Copula	77.8	0.466 ± 0.030	0.473 ± 0.032	+0.007
Census ACS	CTGAN	63.0	0.381 ± 0.063	**0.441 ± 0.096**	**+0.060**
	Gaussian Copula	98.7	0.942 ± 0.009	0.942 ± 0.008	+0.000
	TVAE	92.9	0.665 ± 0.149	0.675 ± 0.136	+0.011
	Vine Copula	76.0	0.483 ± 0.062	**0.697 ± 0.035**	**+0.214**
Credit	CTGAN	29.6	0.447 ± 0.012	0.458 ± 0.020	+0.011
	Gaussian Copula	53.8	0.486 ± 0.030	0.466 ± 0.020	-0.020
	TVAE	92.0	0.438 ± 0.003	0.438 ± 0.003	+0.000
	Vine Copula	43.8	0.439 ± 0.006	0.443 ± 0.006	+0.004
Purchases	CTGAN	18.7	0.375 ± 0.078	**0.519 ± 0.196**	**+0.144**
	Gaussian Copula	61.3	0.977 ± 0.009	0.980 ± 0.022	+0.003
	TVAE	0.2	0.690 ± 0.175	n.d.	-
	Vine Copula	14.5	0.450 ± 0.077	**0.844 ± 0.071**	**+0.394**
Supermarket	CTGAN	0.0	0.343 ± 0.023	n.d.	-
	Gaussian Copula	96.6	0.999 ± 0.001	0.999 ± 0.001	+0.000
	TVAE	0.0	0.766 ± 0.153	n.d.	-
	Vine Copula	0.2	0.458 ± 0.097	n.d.	-

* Full-cohort F1 is always valid and is reported; the filtered F1 is undefined (n.d.) where the retained cohort is empty (0%). Bold Δ F1 values are statistically significant (paired t-test, $p < 0.05$).

To characterize when integrity filtering recovers utility across domains, we tested the paired F1 difference (HIF-filtered minus full synthetic) across all dataset–generator combinations of the cross-architecture benchmark with $N = 10$ seeds. Significant utility recovery occurs in four configurations: Census ACS (ΔF1 = +0.060, 95% CI [+0.020, +0.099], $p = 0.008$ for CTGAN; +0.214, CI [+0.189, +0.238], $p < 0.001$ for Vine Copula) and Online Purchases (ΔF1 = +0.144, CI [+0.034, +0.253], $p = 0.016$ for CTGAN; +0.394, CI [+0.342, +0.447], $p < 0.001$ for Vine Copula). Recovery is selective rather than a deterministic consequence of the violation rate: high-violation cohorts on Credit Default show no significant change despite 70.3% and 56.2% violation rates (CTGAN, ΔF1 = +0.011, $p = 0.15$; Vine Copula, +0.004, $p = 0.06$), and Online Purchases Gaussian Copula shows none at 38.7% (+0.003, $p = 0.61$). Therefore, whether removing flagged records improves utility depends on the strength of the association between the violated conditional structure and the downstream target, not on the violation rate alone. Section 4.4 makes that dependence precise and shows that it is governed by whether the target is one of the auditor's hubs. Conversely, most remaining configurations exhibit small, non-significant changes (e.g., Census ACS Gaussian Copula, ΔF1 = +0.000, $p = 0.90$; Adult TVAE, +0.001, $p = 0.77$; Adult Gaussian Copula, +0.002, $p = 0.77$), confirming that filtering does not penalize well-formed synthetic data. No configuration showed a significant degradation; the nearest was Credit Gaussian Copula (ΔF1 = −0.020, CI [−0.042, +0.002], $p = 0.073$), indicating that pruning can discard flagged records that retain the predictive signal for the target, but this remains a marginal rather than a significant effect. Finally, four transaction configurations (Supermarket Sales CTGAN/TVAE/Vine; Online Purchases TVAE) retained at most $\approx$ 2% of records, so their filtered F1 is not reliably estimable across seeds and their deltas are not reported; their valid full-cohort F1 is given in Table 3. The Online Purchases TVAE cohort retains essentially no records (0.2%), consistent with a cohort whose full-F1 (0.690 $\pm$ 0.175) rests on high-utility but structurally implausible rows; blanket filtering would destroy it, reinforcing the recommendation to reserve exclusionary filtering for applications that tolerate cohort shrinkage (Section 5.1).

Under a Bonferroni correction ($\alpha = 0.05/16 \approx 0.003$), the two Vine Copula recoveries ($p < 0.001$) remained significant, while the Census ACS CTGAN ($p = 0.008$) and Online Purchases CTGAN ($p = 0.016$) fell below the corrected threshold and should be read as suggestive. The aggregate pattern is nevertheless unlikely by chance: four of 16 tests reach $p < 0.05$ versus roughly one expected under a global null, and the effects concentrate on cohorts with the strongest dependency violations.

4.4 The Recovery Effect Is Label-Conditional

The four recoveries above share a property that the violation rate does not explain: in each case, the median-split prediction target is one of the auditor's five selected hubs (`household_income` on Census ACS, `item_total` on Online Purchases). Because a sentinel trained on that hub scores each record by how probable its target value is given the remaining features, low-H records are enriched for rows whose *label* is atypical, not only for rows whose feature geometry is atypical. Pruning them would improve the TSTR F1 through conditional label curation rather than structural repair. We conducted two experiments to determine whether this was the driving force behind the observed effects.

Experiment 1: varying the target at fixed cohort.

We held the target-sweep retained cohorts fixed and swept the downstream target across nine Census ACS columns—the five selected hubs and four non-hubs—so that the filter, retained rows, and retention rate were identical across all conditions and only the label changed (Table 4). Nine of the ten hub target–generator pairs recovered significantly (the sole exception is CTGAN×tenure, $p = 0.109$), with group-mean gains of $+0.077$ (CTGAN) and $+0.112$ (Vine Copula). Seven of the eight non-hub pairs show no detectable effect (the exception is Vine Copula×cost_burden_pct, $p = 0.016$), with group means of $+0.014$ and $+0.010$, an order of magnitude smaller. Among the significant pairs, every hub confidence interval excludes zero, and among the non-hub pairs, only the cost_burden_pct interval does. Because the retained cohort is fixed, this contrast cannot be attributed to the records removed by the auditor; it is entirely a property of the column being predicted.

Notably, `cost_burden_pct` is statistically entangled with two hubs; it correlates 0.72 with $12 \cdot$ `housing_cost/household_income`; however, it shows no recovery under CTGAN ($+0.020$, $p = 0.16$) and only a small marginal one under the Vine Copula ($+0.020$, $p = 0.016$), an order of magnitude below the hub-driven gains. Proximity to an audited hub is insufficient; the target must be audited directly.

Experiment 2: withholding the target from the auditor.

We then re-ran the four recovery settings under two interventions (Table 5): the baseline re-runs the original audits, Target Excluded removes the target from the hub candidate set so that the auditor still sees every column, but no sentinel predicts the target, and Target Hidden removes the target column from the auditor entirely. Retention rises in every arm, as expected, once the target can no longer contribute penalties. Under Target Excluded, the two Census ACS recoveries become non-significant ($p \geq 0.24$) and collapse to within ± 0.019 F1 of zero, while the two Online Purchases configurations are attenuated but retain significance (CTGAN: $+0.021$, $p = 0.027$, an 85% attenuation; Vine: $+0.104$, $p = 0.001$, a 73% attenuation). Under Target Hidden, no configuration retains a significant gain ($p \geq 0.062$), and the two Census ACS effects collapse to within ± 0.007 F1 of the zero. The residual target-excluded effects on Online Purchases indicate that in that dataset, a second pathway contributes—`item_total` participates in arithmetic identities that the NIC and rule layers audit independently of hub selection—so the mechanism is dominant rather than exclusive there.

Interpretation.

Taken together, the two experiments identify a necessary condition rather than a correlate: integrity filtering recovers downstream utility when the auditor models the conditional structure of the target and not otherwise. Therefore, we describe contribution (ii) as *label-conditional curation*. This is a narrower claim than generic structural remediation, and it partially overlaps with conditional label-noise cleaning, a connection that we make explicit in Section 5.2. It is also more actionable: the precondition is the intersection of the auditor's hub set with the intended prediction target, which a practitioner can check from the calibration output alone before committing to a filtering policy and without consulting downstream labels. The diagnostic contribution (i) is unaffected—HIF's detection of dependency ruptures in Sections 4.2 and 4.5 involves no downstream target at all.

Table 4 Target sweep across Census ACS columns at fixed retained cohorts ($N = 10$ seeds). The retention rates are identical in every row (63.3% CTGAN, 76.4% Vine Copula). The integrity filter remains unchanged; only the downstream prediction target varies. Bold ΔF1 values are significant at $p < 0.05$ (paired t-test).

Target	Hub?	CTGAN		Vine Copula	
		ΔF1	p	ΔF1	p
`household_income`	yes	**+0.056**	0.003	**+0.208**	< 0.001
`housing_cost`	yes	**+0.104**	0.002	**+0.114**	< 0.001
`poverty_status`	yes	**+0.091**	0.003	**+0.125**	< 0.001
`education`	yes	**+0.105**	< 0.001	**+0.082**	< 0.001
`tenure`	yes	+0.030	0.109	**+0.030**	0.025
`cost_burden_pct`	no	+0.020	0.16	**+0.020**	0.016
`employment_status`	no	+0.020	0.18	−0.002	0.85
`household_size`	no	+0.020	0.19	+0.014	0.15
`age_group`	no	−0.002	0.77	+0.007	0.45
Hub group mean		+0.077		+0.112	
Non-hub group mean		+0.014		+0.010	

4.5 Comparison with Outlier Detection and Learned-Density Baselines

A natural question is whether standard outlier detection methods, such as Isolation Forest (IF) [5] and Local Outlier Factor (LOF) [6], or a learned-density baseline can serve as alternatives to HIF. Tables 6 and 7 compare the detection performance across five error types on Census ACS at 40% corruption ($N = 10$ seeds); Table 8 replicates the identical protocol on Online Purchases. LOF and IF are configured with a `contamination` parameter set to the known corruption rate (0.4), giving them a tuned decision threshold for F1 in this stress-test setup, whereas HIF uses its fixed non-parametric default frontier, flagging records with $H < 0.5$. As a learned-density baseline, we fit a Gaussian Mixture Model (GMM) on the real data with component count selected by the Bayesian information criterion [44] and score synthetic records by negative log-likelihood; its F1 threshold is the $(1 - 0.4)$ quantile of the real-data scores, the density analog of the oracle `contamination` calibration granted to IF and LOF. For a like-for-like comparison, we additionally report HIF† and GMM†, which flag the top 40% of records by their respective scores, matching the calibration granted to the baselines; in practice, the baselines flag a comparable fraction of records (IF

Table 5 The recovery effect vanishes when the target is withheld from the auditor ($N = 10$ seeds). The experiment re-runs the four configurations under three conditions: baseline (auditor sees the target), Target Excluded (target removed from hub candidates but column visible to auditor), and Target Hidden (target column removed entirely). Retention rates increase as expected once the target can no longer incur penalties. Bold ΔF1 values are significant at $p < 0.05$ (paired t-test).

Configuration	Condition	Retained (%)	ΔF1	p
Census ACS CTGAN	Baseline	62.7	**+0.077**	0.002
	Target Excluded	78.2	−0.019	0.24
	Target Hidden	90.4	−0.003	0.78
Census ACS Vine	Baseline	76.4	**+0.208**	< 0.001
	Target Excluded	88.3	−0.000	0.98
	Target Hidden	93.7	+0.007	0.46
Online Purchases CTGAN	Baseline	19.1	**+0.141**	0.016
	Target Excluded	30.3	+0.021	0.027
	Target Hidden	17.6	+0.007	0.68
Online Purchases Vine	Baseline	13.9	**+0.385**	< 0.001
	Target Excluded	25.5	+0.104	0.001
	Target Hidden	15.9	+0.073	0.062

$\approx 40\%$ on average, 35–46% across error families; LOF $\approx 79\%$, 63–87%; GMM $\approx 72\%$, 60–83%), so the matched HIF operating point is not laxer than theirs.

Under the threshold-independent ROC-AUC on Census ACS, HIF strongly outperforms the geometric baselines on *Conditional Dependency Violations* (ROC-AUC = 0.824, PR-AUC = 0.776), whereas the Isolation Forest is near-random on marginal-preserving dependency swaps (ROC-AUC = 0.514), and LOF achieves lower ranking quality (ROC-AUC = 0.717, PR-AUC = 0.655). A learned-density baseline is competitive with HIF in this family (GMM ROC-AUC = 0.819, PR-AUC = 0.802) but trails HIF at the matched operating point (F1 0.710 vs. 0.719; Table 7). The F1 comparison is markedly more favorable to HIF once operating points are matched: flagging the top 40% of records by HIF penalty yields the highest F1 in three of five error families—F1 = 0.719 on dependency violations (vs. 0.594 for LOF, 0.438 for IF, and 0.710 for GMM†), F1 = 0.542 on feature dropout (vs. 0.521 for LOF), and F1 = 0.790 on random noise injection (vs. 0.616 for LOF)—despite flagging only 40% of records to LOF's $\approx 86\%$. HIF also maintains high sensitivity to random noise (ROC-AUC = 0.890) because breaking marginals simultaneously disrupts conditional dependencies. The PR-AUC results are consistent with the ROC-AUC across the five families (Table 6): HIF leads to dependency violations and feature dropout, and the learned-density baseline is competitive on random noise injection.

Two of the five families are negative controls by construction; therefore, the values in these rows are sanity checks rather than benchmarks. *Row duplication* copies rows drawn from the synthetic cohort itself and perturbs them with $N(0, 0.01\sigma)$ noise on numeric columns, so every planted row is in-distribution and individually valid; detecting it would require a cross-row near-duplicate search, and indeed all four methods sit at chance (ROC-AUC 0.501–0.508 on Census ACS, 0.501–0.511 on Online Purchases). *Covariate shift* replaces rows with verbatim real records sampled from the upper quartile of the first numeric column, which are genuine and *more* plausible than the surrounding synthetic rows, so every plausibility-based scorer lands at or below chance; sub-chance AUC is the correct response to this protocol, not a failure of the method. The apparent wins in these two rows are operating-point artifacts: at the 40% prevalence used throughout, randomly flagging a fraction q of records attains $\mathrm{F1} = 2\pi q/(\pi + q)$ with $\pi = 0.4$, and LOF's row-duplication F1 of 0.535—obtained by flagging 80.3% of records—equals the random expectation of 0.534, just as GMM's 0.544 on Online Purchases (84.3% flag rate) matches its expectation of 0.543. Therefore, the three informative families are random noise injection, feature dropout, and conditional dependency violations.

The second-dataset replication on Online Purchases (Table 8) confirms the pattern against the geometric detectors: HIF leads IF and LOF on dependency violations (ROC-AUC 0.802 vs. 0.491 and 0.584), feature dropout (0.714 vs. 0.287 and 0.613), and random noise (0.829 vs. 0.349 and 0.521)—but exposes a clear boundary of the learned-density baseline: GMM is markedly stronger on this domain. For dependency violations, the GMM attained ROC-AUC 0.954 and PR-AUC 0.908 (vs. 0.802 and 0.639 for HIF) and a matched F1 of 0.890 (vs. 0.668), and it also leads on feature dropout (matched F1 0.737 vs. 0.573) and random noise (0.720 vs. 0.690). We attribute this to the near-deterministic continuous arithmetic identities of the dataset (item_total = item_subtotal + item_tax; item_subtotal = purchase_price $\times$ quantity): violating these constraints moves a row sharply off a low-dimensional manifold that a Gaussian mixture approximates almost exactly, whereas HIF's categorical-centric sentinel ensemble has little structure to exploit (a single categorical column), and its NIC residual thresholds are conservatively calibrated to the natural noise floor. HIF's advantage is therefore concentrated precisely where learned-density baselines are weakest: predominantly continuous spaces whose conditional structure is complex and non-arithmetic (Census ACS: nine numeric features plus a single 51-level categorical, with all five audited hubs continuous), where the one-hot expansion of that categorical into 51 binary dimensions gives the Gaussian mixture a mostly binary support on which to score the sharp continuous dependencies. A caveat: the GMM on Census ACS operates in 60 dimensions (9 numeric + 51 one-hot) with only 2000 training samples, which is severely underpowered for density estimation; its competitive ROC-AUC (0.819 vs. 0.824) may reflect overfitting rather than genuine generalization, and the comparison should be interpreted with this limitation in mind.

Why do standard outlier detectors fail on Dependency Gaps? The Isolation Forest and LOF evaluate the geometric distance or marginal density across the entire continuous feature space. If a synthetic record hallucinates an impossible logic combination (e.g., a 14-year-old with a PhD), where both values are marginally common, the record resides in a dense region of the globally projected feature space. Because the Isolation Forest splits features independently, it rarely isolates the specific interacting dimensions that violate the conditional rule. A learned-density model faces a related but distinct limitation: on Census ACS, it models a 60-dimensional encoding—nine standardized numeric columns joined with 51 one-hot indicators for the sole categorical feature (`state`)—so the sharp conditional structure concentrated in the continuous features is scored through a mostly binary support, which blurs the contrast between a valid record and a dependency-violating one; however, it still comes close to HIF on this family. A Dependency Gap is a structural, semantic error, not a geometric density anomaly, which is why HIF's logic-aware sentinels are competitive with or better than both geometric and learned-density baselines on conditional dependencies over discrete-valued hubs. The replication further shows that this advantage is domain-dependent: when the underlying constraints reduce to continuous arithmetic identities, a learned-density model captures them more sharply than HIF's conservatively calibrated residuals, and the two approaches should be treated as complementary—GMM for near-deterministic continuous constraints, HIF for complex conditional structure over discrete-valued hubs, where it additionally provides per-dependency record attribution that a scalar density score does not.

Table 6 Held-Out Error Detection: ROC-AUC and PR-AUC (Census ACS, 40% corruption). Threshold-independent metrics across ten seeds. HIF strongly outperformed geometric baselines (Isolation Forest, LOF) on conditional dependency violations (ROC-AUC 0.824 vs. 0.514 and 0.717). A learned-density baseline (GMM) was competitive in terms of ranking quality (PR-AUC 0.802 vs. 0.776). The row-duplication and covariate-shift rows are negative controls; random baseline performance is ROC-AUC 0.5, PR-AUC $\approx$ 0.40.

	ROC-AUC				**PR-AUC**			
Error Type	**HIF**	**IF**	**LOF**	**GMM**	**HIF**	**IF**	**LOF**	**GMM**
Random Noise Injection	**0.890**	0.524	0.864	0.874	**0.849**	0.412	0.831	0.848
Row Duplication	0.508	0.504	0.508	0.501	0.408	0.405	0.408	0.404
Feature Dropout	**0.645**	0.315	0.486	0.506	**0.571**	0.295	0.396	0.445
Covariate Shift	0.293	0.469	0.213	0.265	0.355	0.406	0.267	0.280
Conditional Dependency Violations	**0.824**	0.514	0.717	0.819	0.776	0.409	0.655	**0.802**

Table 7 Held-Out Error Detection: F1 at Fixed Operating Points (Census ACS, 40% corruption). HIF† and GMM† flag the top 40% of records by their respective scores, matching oracle `contamination` calibration granted to baselines; unmarked HIF uses the fixed default frontier ($H < 0.5$). At matched operating points, HIF attains the highest F1 on conditional dependency violations (0.719 vs. 0.594 for LOF, 0.710 for GMM†), feature dropout (0.542 vs. 0.521), and random noise injection (0.790 vs. 0.616). For negative-control rows (row duplication, covariate shift), random baseline F1 $\approx$ 0.40 at 40% flag rate.

Error Type	**HIF**	**HIF†**	**IF**	**LOF**	**GMM**	**GMM†**
Random Noise Injection	0.370	**0.790**	0.450	0.616	0.632	0.756
Row Duplication	0.026	0.406	0.423	0.535	0.515	0.402
Feature Dropout	0.072	**0.542**	0.196	0.521	0.492	0.415
Covariate Shift	0.014	0.165	0.387	0.287	0.321	0.221
Conditional Dependency Violations	0.206	**0.719**	0.438	0.594	0.616	0.710

Table 8 Held-Out Error Detection Replication: HIF vs. Baselines (Online Purchases, 40% corruption). Identical protocol and operating point conventions as Tables 6–7. Threshold-independent ROC-AUC and PR-AUC across ten seeds. On this continuously constrained domain, the learned-density baseline was markedly stronger than HIF in ranking quality for *Conditional Dependency Violations* (ROC-AUC 0.954 vs. 0.802; matched F1 0.890 vs. 0.668), while HIF retains an edge over the geometric detectors (IF, LOF) on all three dependency-relevant families. As in Tables 6–7, the row-duplication and covariate-shift rows are negative controls (see text).

	ROC-AUC				PR-AUC			
Error Type	**HIF**	**IF**	**LOF**	**GMM**	**HIF**	**IF**	**LOF**	**GMM**
Random Noise Injection	0.829	0.349	0.521	**0.835**	0.736	0.310	0.458	**0.757**
Row Duplication	0.504	0.501	0.505	0.511	0.409	0.402	0.407	0.409
Feature Dropout	0.714	0.287	0.613	**0.813**	0.570	0.289	0.516	**0.758**
Covariate Shift	0.190	0.294	0.284	0.176	0.341	0.310	0.317	0.258
Conditional Dependency Violations	0.802	0.491	0.584	**0.954**	0.639	0.385	0.479	**0.908**

Table 9 Held-Out Error Detection Replication: F1 at Fixed Operating Points (Online Purchases, 40% corruption). Identical protocol and operating-point conventions as Tables 6–7; the matched operating points mirror the contamination-aware baselines. On this continuously constrained domain, the learned-density baseline remains stronger than HIF on dependency violations, while HIF still leads the geometric detectors on the dependency-relevant families.

Error Type	**HIF**	**HIF**†	**IF**	**LOF**	**GMM**	**GMM**†
Random Noise Injection	0.706	0.690	0.397	0.496	0.608	**0.720**
Feature Dropout	0.624	0.575	0.320	0.544	0.592	**0.737**
Covariate Shift	0.012	0.056	0.325	0.297	0.303	0.124
Row Duplication	0.391	0.398	0.509	0.516	0.544	0.413
Conditional Dependency Violations	0.707	0.668	0.511	0.551	0.613	**0.890**

4.6 Differential Diagnostic: Integrity vs. Loyalty

To determine whether HIF provides unique utility over joint statistical metrics, we evaluated its response at low corruption levels ($p \leq 0.1$) using a conditional-swap protocol whose corrupted rows draw replacement feature values from the real conditional pools (Census ACS, $N = 3$ seeds). At these low rates, the HIF score decays monotonically with p (pooled Spearman $\rho = -0.68$ over the $p \leq 0.1$ levels; from 0.933 at $p = 0$ to 0.913 at $p = 0.1$), whereas the aggregate marginal metrics move negligibly. Moment Matching improves only slightly as replacement values are drawn from real conditional pools (88.56 $\rightarrow$ 88.74), and KS is essentially unchanged (0.927 $\rightarrow$ 0.928). The Joint Correlation Distance moved only marginally (94.6 $\rightarrow$ 94.1, relative shift $< 0.6\%$). The small absolute JCD change reflects dilution: the injected rows ($\leq 10\%$ of the 2,000-row cohort) contribute negligibly to a cohort-averaged correlation statistic. Because HIF emits a per-record penalty, it localizes exactly which rows carry the corrupted conditional relationships, a capability that no aggregate metric provides.

This reading is complementary, not adversarial: under the broad, high-rate permutation protocol described earlier (which destroys pairwise correlations), the JCD decays sharply in lockstep with the HIF score. Therefore, the two tools address related but distinct failure modes: the JCD is a sensitive dataset-level drift detector, whereas HIF provides record-level attribution of conditional dependency violations.

4.7 Component Ablation Study

To understand the contribution of each HIF component, we performed an ablation study on the Census ACS using CTGAN ($N = 5$ seeds; Table 10). Because Census ACS is predominantly continuous—nine numeric columns plus a single 51-level categorical (`state`), with all five audited hubs quantile-binned continuous features—the Logical Sentinel Ensemble (LSE), which scores conditional structure through these hubs, is the most critical component, raising F1 from 0.412 ± 0.032 (no filtering) to 0.837 ± 0.012 while retaining 26.9% of records. In contrast, rule-based filtering flags few records (7.2% filtered out) and leaves utility near baseline, and NIC-only pruning barely changes retention (7.9% filtered out), indicating that the continuous-residual signal NIC thresholds are secondary to the hub-conditional signal in this domain. On Census ACS, NIC operates with only one categorical column (`state`) for context, giving it almost no conditional structure to exploit; its contribution is therefore negligible on this dataset, while it plays a larger role on domains with richer categorical structure (e.g., Online Purchases, Supermarket Sales). Combining LSE with NIC yields intermediate behavior (retention 44.5%, F1 0.672 ± 0.058).

The ablation also illustrates the role of the aggregation method: Full HIF with multiplicative product aggregation ($F1 = 0.490 \pm 0.043$) outperforms the arithmetic mean ($F1 = 0.419 \pm 0.026$). The multiplicative product strongly limits compensation between components, preventing a high score in one dimension from hiding a severe failure in another, pruning an additional 28% of records relative to arithmetic aggregation (retaining 64.0% vs. 91.8%). The magnitude of this advantage should be interpreted with caution because retention and utility are confounded, and the gap is driven primarily by how aggressively each variant is pruned. Consistent with this interpretation, the aggregation choice is immaterial in settings free of structural errors: under Gaussian Copula on Census ACS, where even the most aggressive variant (LSE-only) filters just 4.9% of records, all ablation configurations yield statistically indistinguishable F1 (0.945–0.948).

Aggregation behavior.

The apparent paradox that adding components lowers F1 compared to LSE-only is expected: when NIC and Rules assign near-1.0 scores to a record that LSE penalizes, the geometric mean pulls the overall score up, rescuing borderline records from the $H < 0.5$ threshold. This rescuing effect confirms that the geometric mean is not a strict Logical AND—it is compensatory, but far less so than the arithmetic mean. Thus, the full HIF acts as a balanced filter (64.0% retention, F1 0.490) rather than an ultra-strict one (LSE-only: 26.9%, F1 0.837). The geometric mean is chosen specifically because multiplication limits rescue far more strongly than arithmetic addition.

Table 10 Component Ablation Study on Census ACS (CTGAN, $N = 5$ seeds, Mean ± SEM). The multiplicative product strongly limits compensation between components, pruning more records than the arithmetic mean and reaching the highest F1 score in a cohort with structural errors. In cohorts free of structural errors (Gaussian Copula), all configurations are statistically indistinguishable (F1 0.945–0.948).

Ablation	Retention (%)	F1 (mean ± SEM)	Filtered Out (%)
No filtering	100.0%	0.412 ± 0.032	—
LSE-only	26.9%	0.837 ± 0.012	73.1%
NIC-only	92.1%	0.397 ± 0.026	7.9%
Rules-only	92.8%	0.391 ± 0.022	7.2%
LSE + NIC	44.5%	0.672 ± 0.058	55.5%
Full HIF (arith.)	91.8%	0.419 ± 0.026	8.2%
Full HIF (multiplicative)	**64.0%**	**0.490 ± 0.043**	**36.0%**

5 Discussion

A central finding of our cross-architecture audit (Table 1) is that the Dependency Gap is not restricted to deep neural black-boxes; it appears as a recurring pathology affecting both statistical and neural generative models in distinct ways. In our runs, statistical copulas preserve integrity well on Census ACS and Supermarket Sales (HIF 0.94–0.96 for the Gaussian Copula) but degrade on Online Purchases and Credit Default (HIF 0.49–0.64), and the designed Vine Copula control—despite the highest marginal fit (KS ≥ 0.970)—fails joint conditional tests as designed (HIF 0.03–0.27 on the transaction domains). Conversely, neural models such as CTGAN capture discrete categorical clusters in demographic data (HIF 0.57–0.62) but struggle with continuous arithmetic identities (HIF 0.02 on Supermarket Sales). This reconciles our core motivation: the Dependency Gap does not stem from neural opacity alone or copula rigidity. This is an inherent artifact of standard generative loss functions, which optimize aggregate distributional fit (likelihood, Wasserstein distance, or marginal discrimination) rather than row-level joint conditional integrity. Thus, post-hoc row-level auditing is necessary, regardless of whether the underlying generator is statistical or neural.

5.1 Practical Interpretation, Specificity, and Scalability

A common concern with multiplicative evaluation is that absolute scores appear lower than the near-unity values reported by aggregate 1D distributional metrics [2, 4, 10]. This is intentional. A record with a severe violation of any single dependency should not receive a high integrity score, regardless of how closely the remaining features match the marginal expectations. Under empirical quantile calibration, the false rejection probability for valid tail records is bounded by the Union Bound (Remark 1, Appendix A).

HIF therefore trades absolute scale for dependency specificity: its geometric mean $H(\mathbf{x}) = (\prod_l \mathcal{C}_l)^{1/L}$ requires simultaneous satisfaction across all audited layers, so it prioritizes structural consistency over marginal rarity. This is deliberate in high-stakes settings: a generic anomaly detector sensitive to marginal outliers would still flag valid but rare records, whereas the HIF is calibrated against the real-data confidence floor and emphasizes conditional correctness rather than tail noise.

From a deployment perspective, integrity filtering should be treated as a curation step rather than a universal correction rule. In the strongest recovery settings, the retained cohort falls to roughly 15–19% of the synthetic data, and utility gains are selective. Therefore, we recommend using HIF primarily as a diagnostic tool for generator refinement and reserving exclusionary filtering for applications that can tolerate cohort shrinkage. The target-aware decision rule is straightforward: at the default frontier $H < 0.5$, a utility gain is expected only when the prediction target lies in the selected hub set; otherwise, the filter is typically utility-neutral. Because this check depends only on the real-data calibration, it can be evaluated before any filtering is applied to the data. The caveat is that the condition is necessary but not sufficient, and when the target itself is being curated, a label-noise baseline is a more appropriate comparison.

The LSE audit is also computationally light in the settings studied. Its cost scales linearly with the number of records as $O(N \cdot K)$ across K hubs, and the sweep in Appendix D.1 shows stable performance even at $K = 1$. In our benchmarks, auditing 10,000 synthetic rows required approximately 0.70 s and 100,000 rows required approximately 5.70 s on standard commodity hardware. These measurements concern the tabular regimes studied here; for very wide tables, the Random Forest memory overhead is the likely bottleneck, and sparse-linear or neuro-inspired sentinels may offer a better trade-off between semantic depth and efficiency.

5.2 Limitations and Failure Regimes

The current study has several limitations. First, HIF is not a general covariate-shift detector (Section 4.5), and its advantage over learned density baselines is domain-dependent. For datasets whose constraints reduce to continuous arithmetic identities, a density model can rank violations more sharply than the hub-based sentinel ensemble. Second, hub selection and confidence floor calibration depend on sample size; therefore, sparse regimes may blur the boundary between rare but valid records and genuine violations. Third, the fixed frontier ($H < 0.5$) is practical but not universally optimal; adapting the calibration to non-exchangeable data streams via conformal methods [45] remains open. Fourth, the benchmark suite is limited to static tabular settings and does not cover temporal or relational data. Fifth, the Random Forest sentinel scale with $O(N \cdot K \cdot T)$ may become memory-heavy on very wide tables; lighter or sparse sentinels may be preferable in that regime. Sixth, hub selection via single-variable classification accuracy (Feature Dependency Score $S(h)$) primarily captures univariate dependencies and may miss higher-order interactions (e.g., parity constraints like $A \oplus B \oplus C = 0$) that are not reducible to any single hub's conditional structure; extending hub selection to multi-way interaction measures such as Total Correlation is left to future work.

A more consequential caveat concerns how the utility results should be interpreted. When the prediction target is among the audited hubs (Section 4.4), integrity filtering overlaps with conditional label-noise cleaning; the target-predicting sentinels are, by construction, conditional label models. Therefore, we do not claim that HIF is preferable to purpose-built label-noise methods for that task, and we have not benchmarked it against them. The relevant claim is narrower: HIF audits all hub-level dependencies simultaneously, is calibrated without downstream labels, and provides per-dependency attribution rather than a single label noise score. A direct comparison with conditional label-noise baselines on hub-target configurations is a natural next step.

Finally, HIF should be interpreted jointly with a support-coverage diagnosis. A degenerate generator that emits a single real row repeatedly can score highly under HIF because the row satisfies the learned constraints by construction, even though the cohort has collapsed its support. This is especially relevant for demographic benchmarks, where high HIF values can coexist with low α-precision and limited diversity (Table 1). Therefore, we recommend reporting HIF together with a support-coverage metric such as α-precision, rather than treating the integrity frontier alone as evidence of cohort quality.

5.3 Broader Impact

HIF provides a useful integrity signal for synthetic data in privacy-sensitive domains, such as healthcare, finance, and government statistics. By detecting row-level dependency violations that aggregate metrics misses, practitioners can audit synthetic cohorts before deployment, reducing the risk that downstream analyses are contaminated by structurally invalid records. We recommend using HIF primarily as a diagnostic tool to guide generator refinement by identifying which conditional structures a generator fails to capture rather than as a purely exclusionary filter, since filtering entails a retention–utility trade-off that must be weighed against the needs of the application. When exclusionary filtering is used, the label-conditional curation result (Section 4.4) provides a useful necessary precondition—checking whether the target is among the auditor's hubs—that can be evaluated at calibration time without downstream labels, making the decision transparent and auditable. HIF's open-source release and data-driven constraint discovery lower the barrier to adoption, but practitioners should report HIF jointly with support-coverage diagnostics (α-precision) to avoid over-interpreting high integrity scores as evidence of cohort diversity.

A dual-use consideration: integrity filtering could reduce demographic representation if hubs correlate with sensitive attributes (e.g., income and education). Practitioners should audit representation drift on sensitive attributes before and after filtering, particularly in high-stakes domains.

6 Conclusion

This study introduces the Hybrid Integrity Framework (HIF), a row-level diagnostic tool that detects conditional dependency violations in synthetic tabular data. HIF composes conditional classifiers, spectral-continuity auditing, and mined implication rules into a multiplicative score that strongly limits compensation between components, flagging individual records whose joint configuration violates the learned conditional structure of the real data—violations that aggregate metrics (KS, TVD, JCD) are not designed to expose directly.

Our cross-architecture audit reveals a persistent integrity–fidelity decoupling: generators that match per-column distributions (KS $>$ 0.96) can simultaneously fail

85–100% of dependency integrity tests on constrained domains, and HIF localizes these failures to specific records. We further characterize a label-conditional curation effect: pruning low-HIF records may recover downstream utility when the prediction target is among the auditor's selected hubs, while this benefit largely vanishes otherwise. This yields a practical and checkable condition for when filtering is likely to pay off.

Future work includes extending the HIF to temporal and relational tables, developing conformal calibration for non-exchangeable data streams, and benchmarking against purpose-built label-noise methods on the hub-target configurations identified here. The complete implementation and all experimental configurations are available at https://github.com/ahmed-fouad-lagha/tabular-polygraph.

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Materials availability. Not applicable.

Code availability. The complete open-source implementation, experimental benchmarks, and dataset configurations are available at https://github.com/ahmed-fouad-lagha/tabular-polygraph.

Data availability. The experimental benchmarks and dataset configurations, together with the complete open-source implementation, are available at https://github.com/ahmed-fouad-lagha/tabular-polygraph.

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References

[1] Assefa, S.A., Dervovic, D., Mahfouz, M., Tillman, R.E., Reddy, P., Veloso, M.: Generating synthetic data in finance: Opportunities, challenges and pitfalls. In: Proceedings of the First ACM International Conference on AI in Finance (ICAIF '20), pp. 1–8. ACM, New York, NY, USA (2020). https://doi.org/10.1145/3383455.3422554

[2] Jordon, J., Szpruch, L., Houssiau, F., Bottarelli, M., Cherubin, G., Maple, C., Cohen, S.N., Weller, A.: Synthetic data–what, why and how? arXiv preprint arXiv:2205.03257 (2022)

[3] Tucker, A., Wang, Z., Rotalinti, Y., Myles, P.: Generating high-fidelity synthetic patient data for assessing machine learning healthcare software. NPJ Digital Medicine **3**(147), 1–13 (2020) https://doi.org/10.1038/s41746-020-00353-9

[4] Long, Y., Xu, L., Brintrup, A.: Evaluating inter-column logical relationships in synthetic tabular data generation. arXiv preprint arXiv:2502.04055 (2025)

[5] Liu, F.T., Ting, K.M., Zhou, Z.-H.: Isolation forest. In: 2008 Eighth IEEE International Conference on Data Mining (ICDM), pp. 413–422 (2008). https://doi.org/10.1109/ICDM.2008.17

[6] Breunig, M.M., Kriegel, H.-P., Ng, R.T., Sander, J.: Lof: Identifying density-based local outliers. In: Proceedings of the 2000 ACM SIGMOD International Conference on Management of Data, pp. 93–104 (2000). https://doi.org/10.1145/342009.335388

[7] Sklar, A.: Fonctions de repartition a n dimensions et leurs marges. Publications de l'Institut de Statistique de l'Universite de Paris **8**, 229–231 (1959)

[8] Joe, H.: Families of m-variate distributions with given margins and m(m-1)/2 bivariate dependence parameters. Lecture Notes - Monograph Series **28**, 120–141 (1996) https://doi.org/10.1214/lnms/1215452199

[9] Xu, L., Skoularidou, M., Cuesta-Infante, A., Veeramachaneni, K.: Modeling tabular data using conditional gan. In: Advances in Neural Information Processing Systems, vol. 32 (2019)

[10] Patki, N., Wedge, R., Veeramachaneni, K.: The synthetic data vault. In: IEEE International Conference on Data Science and Advanced Analytics (2016). https://doi.org/10.1109/DSAA.2016.49

[11] Zhao, Z., Kunar, A., Birke, R., Chen, L.Y.: CTAB-GAN: Effective table data synthesizing. In: Proceedings of the 13th Asian Conference on Machine Learning (ACML). Proceedings of Machine Learning Research, vol. 157, pp. 97–112 (2021). PMLR

[12] Kotelnikov, A., Baranchuk, D., Rubachev, I., Babenko, A.: TabDDPM: Modelling tabular data with diffusion models. In: Proceedings of the 40th International Conference on Machine Learning, pp. 17564–17579. PMLR, Honolulu, HI, USA (2023)

[13] Jolicoeur-Martineau, A., Fatras, K., Kachman, T.: Generating and imputing tabular data via diffusion and flow-based gradient-boosted trees. In: International Conference on Artificial Intelligence and Statistics. Proceedings of Machine Learning Research, vol. 238, pp. 1288–1296 (2024). PMLR

[14] Lee, C., Kim, J., Park, N.: Codi: Co-evolving contrastive diffusion models for mixed-type tabular synthesis. In: Proceedings of the 40th International Conference on Machine Learning (ICML 2023). Proceedings of Machine Learning Research, vol. 202, pp. 18940–18956 (2023)

[15] Zhu, Y., Zhao, Z., Birke, R., Chen, L.Y.: Permutation-invariant tabular data synthesis. arXiv preprint arXiv:2211.09286 (2022)

[16] Zhao, Z., Birke, R., Kunar, A., Chen, L.Y.: Fed-tgan: Federated learning framework for synthesizing tabular data. arXiv preprint arXiv:2108.07927 (2021)

[17] Yu, K., et al.: Shap distance: An explainability-aware metric for evaluating the semantic fidelity of synthetic tabular data. arXiv preprint arXiv:2511.17590 (2025)

[18] Valiant, P., Valiant, G.: An automatic inequality prover and instance optimal identity testing. SIAM Journal on Computing **43**(1), 156–175 (2014)

[19] Pensia, A., Jog, V., Loh, P.-L.: The sample complexity of simple binary hypothesis testing. In: Proceedings of the 37th Conference on Learning Theory (COLT) (2024)

[20] Paninski, L.: A coincidence-based test for uniformity given very sparsely sampled discrete data. IEEE Transactions on Information Theory **54**(10), 4750–4755 (2008) https://doi.org/10.1109/TIT.2008.928987

[21] Bhattacharyya, A., Gayen, S., Kandasamy, S., Vinodchandran, N.V.: Testing product distributions: A closer look. In: Proceedings of the 32nd International Conference on Algorithmic Learning Theory (ALT). Proceedings of Machine Learning Research, vol. 132, pp. 367–396 (2021). PMLR

[22] Neykov, M., Balakrishnan, S., Wasserman, L.: Minimax optimal conditional independence testing. The Annals of Statistics **49**(4), 2151–2177 (2021) https://doi.org/10.1214/20-AOS2030

[23] Umesh, C., Schultz, K., Mahendra, M., Bej, S., Wolkenhauer, O.: Preserving logical and functional dependencies in synthetic tabular data. Pattern Recognition **163**, 111459 (2025) https://doi.org/10.1016/j.patcog.2025.111459

[24] Alaa, A., Breugel, B., Saveliev, E.S., Schaar, M.: How faithful is your synthetic data? sample-level metrics for evaluating and auditing generative models. In: Proceedings of the 39th International Conference on Machine Learning (ICML 2022). Proceedings of Machine Learning Research, vol. 162, pp. 290–306 (2022)

[25] Seedat, N., Crabbé, J., Schaar, M.: Data-SUITE: Data-centric identification of in-distribution incongruous examples. In: Proceedings of the 39th International Conference on Machine Learning (ICML 2022). Proceedings of Machine Learning Research, vol. 162, pp. 19467–19496 (2022)

[26] Breugel, B., Sun, H., Qian, Z., Schaar, M.: Membership inference attacks against synthetic data through overfitting detection. In: Proceedings of the 26th International Conference on Artificial Intelligence and Statistics (AISTATS 2023).

Proceedings of Machine Learning Research, vol. 206, pp. 3493–3514 (2023)

[27] Jiang, X., Simidjievski, N., Jamnik, M.: Tabstruct: Measuring structural fidelity of tabular data. arXiv preprint arXiv:2509.11950 (2025) https://doi.org/10.48550/arxiv.2509.11950

[28] Seedat, N., Huynh, N., Breugel, B., Schaar, M.: Curated LLM: Synergy of LLMs and data curation for tabular augmentation in low-data regimes. In: Proceedings of the 41st International Conference on Machine Learning (ICML 2024). Proceedings of Machine Learning Research, vol. 235, pp. 44060–44092 (2024)

[29] Huhtala, Y., Kärkkäinen, J., Porkka, P., Toivonen, H.: TANE: An efficient algorithm for discovering functional and approximate dependencies. The Computer Journal **42**(2), 100–111 (1999)

[30] Chu, X., Ilyas, I.F., Papotti, P.: Discovering denial constraints. In: Proceedings of the VLDB Endowment (PVLDB), vol. 6(13), pp. 1498–1509 (2013)

[31] Fefferman, C., Mitter, S., Narayanan, H.: Testing the manifold hypothesis. Journal of the American Mathematical Society **29**(4), 983–1049 (2016) https://doi.org/10.1090/jams/852

[32] Breiman, L.: Random forests. Machine learning **45**, 5–32 (2001)

[33] Friedman, J.H., Popescu, B.E.: Predictive learning via rule ensembles. The Annals of Applied Statistics **2**(3), 916–954 (2008) https://doi.org/10.1214/07-AOAS148

[34] Shafer, G., Vovk, V.: A tutorial on conformal prediction. Journal of Machine Learning Research **9**(3), 371–421 (2008)

[35] Vovk, V., Gammerman, A., Shafer, G.: Algorithmic Learning in a Random World. Springer, New York, NY (2005)

[36] Angelopoulos, A.N., Bates, S.: Conformal prediction: A gentle introduction. Foundations and Trends in Machine Learning **16**(4), 494–591 (2023) https://doi.org/10.1561/2200000101

[37] Abdi, H., Williams, L.J.: Principal component analysis. Wiley Interdisciplinary Reviews: Computational Statistics **2**(4), 433–459 (2010)

[38] Greenacre, M.: Correspondence Analysis in Practice. Chapman and Hall/CRC, Boca Raton, FL (2007)

[39] Friedman, J.H.: Greedy function approximation: A gradient boosting machine. The Annals of Statistics **29**(5), 1189–1232 (2001) https://doi.org/10.1214/aos/1013203451

[40] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu, T.-Y.: Lightgbm: A highly efficient gradient boosting decision tree. In: Advances in Neural Information Processing Systems, vol. 30, pp. 3146–3154 (2017)

[41] Tukey, J.W.: Exploratory Data Analysis. Addison-Wesley, Reading, MA (1977)

[42] Jordon, J., Yoon, J., Schaar, M.: Pate-gan: Generating synthetic data with differential privacy guarantees. In: International Conference on Learning Representations (2019)

[43] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., *et al.*: Scikit-learn: Machine learning in Python. Journal of Machine Learning Research **12**, 2825–2830 (2011)

[44] Schwarz, G.: Estimating the dimension of a model. The Annals of Statistics **6**(2), 461–464 (1978) https://doi.org/10.1214/aos/1176344136

[45] Barber, R.F., Candès, E.J., Ramdas, A., Tibshirani, R.J.: Conformal prediction beyond exchangeability. The Annals of Statistics **51**(2), 816–845 (2023) https://doi.org/10.1214/23-AOS2276

[46] Karabulut, E., Groth, P., Degeler, V.: Neurosymbolic association rule mining from tabular data. In: Proceedings of Machine Learning Research (PMLR), vol. 284, pp. 565–588 (2025)

[47] Lundberg, S.M., Erion, G., Chen, H., DeGrave, A., Prutkin, J.M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N., Lee, S.-I.: From local explanations to global understanding with explainable ai for trees. Nature Machine Intelligence **2**(1), 56–67 (2020) https://doi.org/10.1038/s42256-019-0138-9

Appendix A Formal Results for the HIF Score Under Stated Assumptions

We provide a formal analysis of the Hybrid Integrity Framework (HIF) score properties under the assumptions stated below. The results should be interpreted as assumption-scoped statements rather than as unconditional guarantees.

None of the statements below guarantee conformal coverage. The confidence floor δ_h is the empirical 5th percentile of each sentinel's out-of-bag conditional probabilities, not a conformal quantile computed on an exchangeable calibration set: out-of-bag predictions are not exchangeable with test points in the required sense, and a per-hub *marginal* quantile does not yield a *per-record* bound. HIF therefore provides measured—not guaranteed—operating characteristics; on genuine held-out real data, its false-rejection (violation) rate ranges from 0.0% (Supermarket Sales) to 14.0% (Credit Default) across datasets (Section 4).

Proposition 1 (Intersection Soundness). The HIF score $H(\mathbf{x})$ defines a membership certificate for the intersection of semantic manifolds $\mathcal{M}_{total} = \bigcap_l \mathcal{M}_l$. Under the idealization that each component validity satisfies $\mathcal{C}_l(\mathbf{x}) = 1$ iff $\mathbf{x} \in \mathcal{M}_l$ and $\mathcal{C}_l(\mathbf{x}) \in [0, 1)$ otherwise, $H(\mathbf{x}) = \left(\prod_l \mathcal{C}_l(\mathbf{x})\right)^{1/L}$ attains its maximum value 1 exactly on $\mathcal{M}_{total}$:

$$\nVdash[H(\mathbf{x}) = 1] = \prod_l \nVdash[\mathbf{x} \in \mathcal{M}_l] \tag{A1}$$

Proof. *Completeness:* if $\mathbf{x} \in \mathcal{M}_{total}$ then $\mathcal{C}_l(\mathbf{x}) = 1$ for all l, so $H(\mathbf{x}) = 1$. *Soundness (non-compensatory property):* if $\mathbf{x} \notin \mathcal{M}_{total}$, there exists l^* with $\mathcal{C}_{l^*}(\mathbf{x}) < 1$, and since all validities lie in $[0, 1]$, the geometric mean satisfies $H(\mathbf{x}) < 1$. *Strict convergence:* an "Absolute Violation" in any layer ($\mathcal{P}_{l^*} \to 1$) drives $H(\mathbf{x}) \to 0$ regardless of other layers. $H(\mathbf{x})$ behaves as a conservative diagnostic for the joint satisfaction of latent manifold constraints, matching the implementation in Algorithm 3.

Proposition 2 (Asymptotic Convergence of the HIF Penalty). Let $\hat{P}_N(x_h \mid \mathbf{x}_{\setminus h})$ be the empirical conditional probability estimator (Sentinel f_h) trained on a ground-truth dataset $\mathcal{D}_{real}$ of size N. Assuming the base estimator is statistically consistent (e.g., consistent non-parametric trees), as $N \to \infty$, $\hat{P}_N$ converges in probability to the true conditional distribution $P^*(x_h \mid \mathbf{x}_{\setminus h})$, and the empirical penalty $\mathcal{P}_h^{(N)}(\mathbf{x})$ converges pointwise in probability to the corresponding population penalty:

$$\lim_{N \to \infty} P\left(\left|\mathcal{P}_h^{(N)}(\mathbf{x}) - \mathcal{P}_h^*(\mathbf{x})\right| > \epsilon\right) = 0 \quad \forall \epsilon > 0,\ \mathbf{x} \in \mathcal{M}_{real} \tag{A2}$$

Proof sketch. *(i)* Because the LSE leverages consistent non-parametric estimators, the empirical conditional probabilities converge pointwise in probability to the true Bayesian posterior P^* (provided the feature space is bounded and leaf nodes grow sub-linearly with N); the pointwise statement reflects that sup-norm convergence is not generally guaranteed for forest-based estimators. *(ii)* Defining the confidence floor δ_h as the empirical 5th-percentile of the sentinel's out-of-bag probabilities yields an asymptotic upper bound of approximately $\alpha = 0.05$ on false positives under calibration assumptions. *(iii)* Records penalized by the converged HIF ($\mathcal{P}^* > 0$) are expected to lie in lower-probability conditional regions of the manifold, supporting a structural-violation interpretation. *(iv)* By the Uniform Law of Large Numbers and the Continuous Mapping Theorem, the empirical Feature Dependency Score $S^{(N)}(h)$ converges in probability to the Bayes optimal classification accuracy $S^*(h)$, implying asymptotically stable hub ranking.

Remark 1 (Type I Error Upper Bound & Cohort-Purity Approximation). Let $\mathbf{x} \sim P_{real}$ be a valid record on the ground-truth manifold $\mathcal{M}_{real}$ (including rare tail outliers). Under quantile calibration ($\delta_h = Q_\alpha$ and continuous tolerance bandwidth $\gamma_y = Q_{1-\beta}$), the false rejection probability is upper-bounded by the Union Bound:

$$P\left(H(\mathbf{x}) < 0.5 \mid \mathbf{x} \in \mathcal{M}_{real}\right) \leq \sum_{h \in \mathcal{H}} \alpha_h + \sum_y \beta_y + \gamma_{rule} \tag{A3}$$

where γ_{rule} is the rule-layer false-rejection rate of valid records under hard application of the mined implication rules (Section 3); it is nonzero in practice because each rule holds on only $\geq 95\%$ of real records by construction, and its empirical counterpart is the held-out violation rate quantified at the end of this remark. Furthermore, let a generative model produce a noisy mixture $P_{syn} = (1-\eta)P_{real} + \eta P_{hallucinated}$, where $\eta \in [0,1]$ is the structural hallucination rate of the generative model. Rejection filtering with a threshold $H \geq 0.5$ yields an approximate retained-cohort purity of:

$$\text{Purity}(H \geq 0.5) = \frac{(1-\eta)(1-\alpha)}{(1-\eta)(1-\alpha) + \eta\beta_{hallucination}} \tag{A4}$$

where $\beta_{hallucination}$ is the leakage probability of hallucinated records past the filter. Equation (A4) is a stylized Bayes calculation that illustrates the mechanism by which filtering can raise retained-cohort purity; however, in practice, α, β, and η are unknown and must be estimated empirically rather than assumed. The measured counterpart is the false rejection (violation) rate of the HIF on genuine held-out real data, which ranges from 0.0% (Supermarket Sales), 1.3% (Census ACS), and 4.3% (Adult) to 6.8% (Online Purchases) and 14.0% (Credit Default)—see Section 4. Therefore, the framework is best read as a diagnostic tool whose operating point is established empirically, not as one carrying guaranteed error bounds.

Appendix B End-to-End Auditing Procedure

Algorithm 2 summarizes the complete manifold calibration procedure (hub selection, confidence floor calibration, NIC regressor training, and rule mining), and Algorithm 3 details the end-to-end auditing procedure for scoring a synthetic dataset with the trained HIF components.

Algorithm 2 HIF Manifold Calibration

Input: Ground-truth data $\mathcal{D}_{real}$, Hub count K_{hub}
Output: Sentinels $\{f_h\}$, Floors $\{\delta_h\}$, NIC Regressors $\{r_y\}$, Thresholds $\{\tau_y\}$
$\mathcal{H} \leftarrow$ Select top K_{hub} features by Feature Dependency Score $S(h)$
for each hub $h \in \mathcal{H}$ **do**
  Train Sentinel $f_h : \mathcal{X}_{\setminus h} \to [0,1]^{|\mathcal{X}_h|}$ on $\mathcal{D}_{real}$ to predict x_h
  $\delta_h \leftarrow \max(\text{Percentile}_5(\{f_h^{\text{OOB}}(\mathbf{x}_h \mid \mathbf{x}_{\setminus h}) \mid \mathbf{x} \in \mathcal{D}_{real}\}), 0.01)$
end for
$\mathcal{Z} \leftarrow \text{SVD}(\text{Scale}(\text{OneHot}(\mathcal{D}_{cat})))$ ▷ Non-Centered Spectral Manifold
for each continuous feature y **do**
  Train Gradient Boosting regressor $r_y : \mathcal{Z} \to \mathbb{R}$ on $\mathcal{D}_{real}$
  $\mathcal{E}_y \leftarrow \{|y - r_y(\mathbf{z})| \mid \mathbf{x} \in \mathcal{D}_{real}\}$ ▷ Ground-truth residuals
  $\tau_y \leftarrow \max(Q_{0.95}(\mathcal{E}_y), \text{Median}(\mathcal{E}_y) + 2 \cdot \text{MAD}(\mathcal{E}_y))$ ▷ Robust Threshold
  $\gamma_y \leftarrow \max(Q_{0.98}(\mathcal{E}_y), 2 \cdot \tau_y, 3 \cdot \text{MAD}(\mathcal{E}_y), 0.1)$ ▷ Dynamic Gamma Scaling
end for
$\mathcal{R} \leftarrow \text{MineImplicationRules}(\mathcal{D}_{real})$ ▷ Apriori-style, min_conf=0.95, min_supp=0.005

Algorithm 3 HIF Synthetic Data Audit

Input: Synthetic record $\mathbf{x}$, Trained components from Alg 2
Output: Record Integrity Score $H(\mathbf{x})$

Step 1: Logical Sentinel Audit (Categorical)
for each hub $h \in \mathcal{H}$ **do**
  $\mathcal{P}_h \leftarrow \text{clip}\left(\frac{\delta_h - f_h(\mathbf{x}_h|\mathbf{x}_{\setminus h})}{\delta_h}, 0, 1\right)$
end for
$\mathcal{P}_{cat} \leftarrow 1 - \prod_{h \in \mathcal{H}}(1 - \mathcal{P}_h)$ ▷ Multiplicative violation aggregation
Step 2: Neighbor-Invariant Continuity Audit (Continuous)
$\mathbf{z} \leftarrow \phi(\mathbf{x}_{cat})$ ▷ Apply learned standardized projection
for each continuous feature y **do**
  $\rho_y \leftarrow |y - r_y(\mathbf{z})|$
  $\mathcal{P}_y \leftarrow \text{clip}\left(\frac{\rho_y - \tau_y}{\gamma_y}, 0, 1\right)$
end for
$\mathcal{P}_{cont} \leftarrow \max_y \mathcal{P}_y$
Step 3: Integrity Aggregation
$\mathcal{P}_{rule} \leftarrow 1$ if $\mathbf{x}$ violates any $r \in \mathcal{R}$ else 0
$H(\mathbf{x}) \leftarrow \left(\prod_l (1 - \mathcal{P}_l)\right)^{1/L} \in [0, 1]$ ▷ Geometric mean of validities

Appendix C Logical Sentinel Ensemble (LSE) Implementation Details

On the U.S. Census ACS dataset, the Feature Dependency Score $S(h)$ identified the following top-5 hubs: `household_income`, `housing_cost`, `poverty_status`, `education`, and `tenure`. The confidence floor δ_h is the 5th-percentile of the *out-of-bag* probability mass that a sentinel assigns to the correct category in the ground-truth data (safety floor 0.01). Out-of-bag predictions yield out-of-sample floors without a held-out split, reflecting generalization rather than memorized fit, and avoid the instability of the 1st-percentile thresholds in sparse data [4, 46]. Hub selection is deterministic: sentinel discovery uses a fixed random state with non-shuffled cross-validation folds, so the hub set depends only on the ground-truth data and is verified to be identical across all $N = 10$ held-out seeds for every benchmark dataset (e.g., `household_income` through `tenure` on Census ACS; `quantity` through `item_tax` on Online Purchases).

Appendix D Extended Results and Sensitivity Analysis

D.1 Hyperparameter Sensitivity

We evaluated HIF robustness across its three key hyperparameters on the Census ACS dataset (2,000 records with 5% injected violations; $N = 5$ seeds, mean $\pm$ SEM). The continuous HIF score is the primary metric, and the violation rate is derived by applying the violation threshold to row-level penalties. The HIF score is stable across hub counts (varying by $< 1\%$ from 0.978 ± 0.0004 at $K = 1$ to 0.987 ± 0.0008 at

$K \geq 10$), since geometric mean aggregation prevents added sentinels from diluting the score; the hub set saturates at $K = 10$ because Census ACS contains only 10 features. Scores were likewise stable for confidence floor percentiles 1–5 (0.9848 ± 0.0003 at percentile 1 vs. 0.9840 ± 0.0003 at percentile 5; minor 0.3% drop to 0.9815 ± 0.0004 at the 10th), making the 5th-percentile default a good sensitivity–robustness balance. Finally, the continuous score is invariant to the binary frontier (0.984 ± 0.0003 at every setting, since the frontier is applied only after scoring), while the violation rate decreases monotonically with the frontier (0.82% at $H < 0.3$, 0.43% at $H < 0.5$, 0.25% at $H < 0.7$), allowing practitioners to tune the operating point without retraining. The same frontier sweep extends to the downstream utility protocol (Section 4.3): the integrity-filtered F1 gain on Census ACS CTGAN is significant at the strict and default frontiers and grows monotonically with strictness (Table D1).

Table D1 Downstream-Utility Threshold Sensitivity (Census ACS, CTGAN, $N = 10$ seeds, mean). Integrity-filtered TSTR F1 and retention at each retention frontier; records with H below the frontier are pruned, so a higher frontier is stricter. The unfiltered cohort achieves F1 = 0.381. The recovery effect is significant at the strict and default frontiers ($H \geq 0.5$) and grows with filtering strictness, but vanishes at the loose frontier.

Retention frontier H	Retained	F1	ΔF1	Paired t p	Wilcoxon p	95% CI for ΔF1
$H \geq 0.7$ (strict)	30.0%	0.761	+0.380	< 0.001	0.002	[+0.332, +0.428]
$H \geq 0.5$ (default)	63.0%	0.441	+0.060	0.008	0.002	[+0.020, +0.099]
$H \geq 0.3$ (loose)	88.5%	0.382	+0.001	0.92	0.74	[−0.025, +0.028]

The strict frontier ($H \geq 0.7$) achieves a large F1 gain (+0.380) but retains only 30% of the synthetic cohort, which may be impractical for applications requiring sufficient training data. The default frontier ($H \geq 0.5$) offers a more balanced tradeoff (63% retention, +0.060 F1), while the loose frontier ($H \geq 0.3$) provides negligible utility gain. Practitioners should choose the operating point based on their tolerance for cohort shrinkage versus utility recovery needs; the threshold can be tuned without retraining the auditor.

D.2 Sample-Complexity Floor at HIF's Operating Configuration

These bounds are computable in the HIF operating configuration. The identity/uniformity-testing bound $m \gtrsim \sqrt{S}/\varepsilon^2$ [18, 20] inverted at the per-hub, per-category sample sizes HIF actually conditions on yields per-cell detectability floors $\varepsilon_{\text{cell}} = (S_{\text{cell}}/n_{\text{cell}}^2)^{1/4}$, where n_{cell} is the number of real rows with hub value c and S_{cell} is the distinct joint configuration of the non-hub features in that cell. Table D2 reports these values for each benchmark dataset using the auditor's default configuration (five hubs and ten quantile bins for continuous features). The cohort-level analog $\varepsilon_{\text{joint}} = (S_{\text{joint}}/n^2)^{1/4}$ with S_{joint} distinct full-row configurations and n the dataset's own cohort size (2000 for Census ACS, Adult, and Credit; 664 for Online Purchases; 1000 for Supermarket Sales) is also shown.

Table D2 Sample-Complexity Floor: per-cell detectability $\varepsilon_{\text{cell}}$ inverted from the identity-testing bound at HIF's actual hub-conditioned sample sizes (5 hubs per dataset, 10 quantile bins). The worst cell (smallest n_{cell}) drives the floor; aggregate $\varepsilon_{\text{joint}}$ assumes a global uniformity test on the full joint.

Dataset	Rows	Med. n_{cell}	Min n_{cell}	Med. $\varepsilon_{\text{cell}}$	Worst $\varepsilon_{\text{cell}}$	$\varepsilon_{\text{joint}}$
Census ACS	2000	200	198	0.266	0.267	0.150
Adult	2000	5.5	1	0.654	1.000	0.149
Credit	2000	12	1	0.537	1.000	0.150
Online Purchases	664	66	1	0.325	1.000	0.193
Supermarket Sales	1000	100	100	0.316	0.316	0.178

The worst-case floors ($\varepsilon \approx 1.0$) occur when long-tail hub categories leave cells with $n_{\text{cell}} \leq 12$ (Adult `native_country`, Credit `pay_*`, Online Purchases `quantity`). Census ACS, whose hubs (`household_income`, `housing_cost`, `poverty_status`, `education`, `tenure`) all have balanced category counts ($n_{\text{cell}} \geq 198$), achieves $\varepsilon \approx 0.27$; Supermarket Sales' hubs are likewise balanced ($n_{\text{cell}} = 100$, $\varepsilon \approx 0.32$). The cohort-level analog $\varepsilon_{\text{joint}} \approx 0.15$–$0.19$ is tighter because the full sample $n = 2000$ is used without splitting, but it delivers no per-record attribution. HIF's row-level design therefore trades per-cell detectability for granularity; the multi-hub geometric aggregation and the confidence-floor safety minimum (0.01) are the design elements that dampen the influence of long-tail cells on the final score.

D.3 Scalability and Baseline Comparisons

We measured HIF's auditing cost directly and compared it analytically—not empirically—against the **SHAP Distance** semantic fidelity metric [17], a representative explainability-aware evaluation baseline; we did not run SHAP Distance ourselves, so no accuracy comparison between the two is claimed here. HIF scoring scales linearly with the number of audited records, $O(N \cdot K)$, where K is the number of hubs. In our empirical scalability benchmark, fitting the 5-hub LSE auditor on 2,000 real records took 3.2 s, after which scoring 10,000 synthetic records took 0.70 ± 0.07 s and scoring 100,000 records took 5.70 ± 0.10 s on commodity multi-core hardware (fit amortized once; wall-clock averaged over three repeats). The marginal per-10,000-row cost at the 100,000-row scale ($5.70\,/\,10 \approx 0.57$ s per 10k rows) confirms the linear scaling. SHAP-based auditing carries a per-record model-explanation cost rather than a single forward pass: exact Shapley values over D features require $O(2^D)$ coalitions in general, and although TreeSHAP computes them exactly in polynomial time for tree ensembles [47], the per-record cost still scales with the ensemble size and depth. Therefore, HIF's per-record cost is substantially lower at fixed D, which is the basis on which we prefer it for repeated auditing at scale; establishing whether it also matches SHAP Distance on detection accuracy would require a direct empirical comparison that we leave to future work.